



Higher-order integration of index-3 DAE with friction using time finite elements on Lie groups

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Abstract

In multibody dynamics, Euler angles and Euler parameters (unit-quaternions) are common choices of global orientation parameters. Both approaches have their limitations, e.g., gimbal lock and an extra normalization constraint, respectively. To circumvent these, the Lie-group based frameworks methods employ $SO(3)$ or $SE(3)$ representation of rotations with corresponding higher order Lie-group integration methods. However, quaternions represent a much more compact representation of rotations and offer a computational advantage over $SO(3)$ or $SE(3)$ representations. Implicit integration inevitably requires computation of Jacobian matrices for Newton-Raphson iterations. However, deriving the component of Newton-Raphson matrix corresponding to body-frame rotations within $SO(3)$ formalism is not limited to calculation of partial derivatives. In fact, the process of deriving these matrices involves a sequence of algebraic manipulations. Subsequently, they often need to be derived for the acting forces and torques, on a case-by-case basis. It is this limitation that emerges as a bottle-neck for modeling friction within Lie-group formalism, as friction loads are intricate nonlinear functions of generalized coordinates (and hence quaternions), velocities as well as Lagrange multipliers. To address this limitation, we propose a systematic partial-derivative based approach for computing the sub-element coefficient matrices with respect to body-frame Euler-rotations. This is achieved by defining general-purpose operators, as presented in this work, that support convenient computer application due to accessibility to a wide range of numerical and auto-differentiation tools. Further, recent advances in time-finite element based methods have demonstrated higher-order accurate solutions for nonlinear and chaotic systems with relatively simple computer implementation. This work implements a family of time-finite element integrators in conjunction with exponential integration of rotations using Lie-group properties of quaternions. Further, numerical experiments showed that the Lie-group time-finite element integrators compare favorably with Lie-group Newmark- β and trapezoidal schemes.

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1 Introduction

In the multibody formulations based on global-coordinates, spatial locations are defined by Cartesian coordinates and global orientations are defined by unit quaternions or Euler-angles. In such cases, carrying out higher-order fully-implicit integration simply requires evaluation of Jacobian matrices, i.e., the partial derivatives with respect to generalized coordinates. However, simulating multibody systems in absolute coordinates, where spatial positions are defined by Cartesian coordinates and pose is defined by local rotations (also called local integration coordinate), implicit integration requires evaluation of Jacobian matrix (or equivalents) with respect to the body-frame Euler-rotation vector. Nevertheless, the body-frame Euler-rotations are not generalized coordinates, and hence can not be used for evaluating Jacobians. Instead, the recent work by Kissel et al. [1–3] rely on quantifying the slight change in the vector of interest by evaluating the small change in the rotation matrix, resulting in a matrix-valued operator Π . This provides a mapping between small changes in rotation matrices, and those in orientations in body-fixed frame. This contribution is important as the resulting first order sensitivities of kinematic constraints and constraint reaction forces are vital in the context of implicit integration. In addition, the authors reported significant improvement in computational performance with this approach, when compared to the methods employing quaternions or Euler-angles. Moreover, these formulations are described in standard matrix notation that aim to introduce the mainstream engineering community to powerful, yet abstract, Lie-group formulations. Although these contributions also laid foundations for exploring higher-order integration, one important limitation that was reported, was exclusion of friction from simulation. We observed that this limitation also arises because evaluation of operator Π for complicated nonlinear functions requires explicit definition.

For body-fixed constant vectors such as body-frame location vectors, evaluating these coefficient matrices is relatively straightforward [1, 2]. However, for nonlinear and time-varying vectors such as forces and torques, the process involves taking time-derivatives and algebraic manipulations to evaluate the coefficient matrices on a case-by-case basis. When the acting forces are represented by complex nonlinear functions of generalized coordinates, for instance friction forces [4–6], analytical evaluation of respective coefficient matrices can be tedious and often impossible for efficient computer applications. Due to this limitation, the extending absolute-coordinate formulation, in its current form, to account for friction, or any arbitrary functions of nonlinear forces dependent on generalized coordinates, can be difficult.

The absolute coordinate formulation [1–3] implements first-order Lie-integration on $SO(3)$ rotation matrices, and emphasizes on the scope of developing the formulation further, for higher-order implicit integration. The existing literature on higher order implicit Lie-group integrators includes Lie-group generalized α methods that are often limited to second-order accuracy [7, 8]. Furthermore, Munthe-Kaas Runge-Kutta methods [9, 10] are widely acclaimed for being A-stable even in the explicit form, and have been employed for engineering applications [11]. Weiloch et al. [12] have further contributed second, third and fourth order accurate BDF integrators on Lie groups. In addition, the authors, in [12], claim that BDF methods feature reduced number of Lie-braket evaluations and hence are more efficient than Munthe-Kaas methods. However, BDF integration methods have been

demonstrated on $\mathbb{R}^3 \times \text{SO}(3)$ and $\text{SE}(3)$ formalisms. The $\text{SE}(3)$ group is the most natural and unified representation of a rigid body pose and motion. It treats the entire 6-DOF motion as an element of a single Lie group. In the context of constrained dynamics, the constraints can be formulated directly on the full six dimensional state space or its velocity space, leveraging powerful tools from differential geometry and Lie group theory [13]. The $\mathbb{R}^3 \times \text{SO}(3)$ representation, on the other hand, separates position and orientation, which is intuitive for many applications. However, rotation matrices must maintain their orthogonality and unit-determinant properties (9 elements with 6 constraints, leaving 3 DOF). While powerful for standard kinematics, managing these implicit constraints during numerical integration or when imposing additional motion constraints can be computationally intensive.

The $\mathbb{R}^3 \times S^3$ representation of rigid body motion offers the most compact singularity-free representation for rotation using quaternions, which is a major advantage over minimal parameterizations like Euler angles. It is often preferred in simulations and control due to its numerical stability. The main trade-off is the need to explicitly enforce the unit-norm constraint of the quaternion (4 elements with 1 constraint, leaving 3 DOF) during calculations, which adds an algebraic constraint to the dynamics equations. Furthermore, Terze et al. proposed redundancy-free Lie-integration in minimal form [11, 14], by circumventing the normalization constraint using Lie-group properties of unit quaternion group $S^3 = \{\mathbf{p} \in \mathbb{R}^4 \mid \mathbf{p}^T \mathbf{p} = 1\}$. In fact, for quaternions, the exponential map can be written in an efficient closed form and numerically efficient truncations for inverse of differential of exponential map can be performed as well [9]. For these listed advantages, it is worth utilizing powerful properties of quaternion Lie-groups to develop implicit integrators using this approach. Recently, [4, 5] used first-order exponential integration on S^3 group for a unified formalism for smooth and non-smooth dynamics problems, in optimization framework. Furthermore, the availability of exact analytical expression for matrix exponential [4, 5, 11, 14] assists in saving cost on Lie-bracket evaluations.

1.1 Review of time finite elements

Time finite element (TFE) methods based on Hamilton's least action principle were first introduced by Bailey [15] and Liepholtz [16]. In later years of that decade, Borri et al. developed continuous TFE methods for multibody systems with large rotations, based on $\text{SE}(3)$ representation of positions and rotations [17–19]. These contributions introduced evaluation of TFE discretized tangent matrices separately for each individual constraint and assembling them later as diagonal entries. Subsequently, in addition to tedious computer implementation, the resulting dimensionality of equations is larger than in joint-coordinate formulation.

In 1990s, discontinuous Galerkin methods were developed by Hulbert and Hughes, applying space-time finite element formulations to structural dynamics problems using classical polynomial interpolation schemes in both space and time [20, 21]. These methods were based on Hamilton's single-field and two-field principles and demonstrated high accuracy with unconditionally stable time integration schemes exhibiting asymptotic annihilation.

In most classical integration methods, the direct integration approaches such as Newmark- β , generalized- α are preferred as they carry advantages of relatively easier computer implementation, in contrast to Runge-Kutta methods that require state-space conversions. A continuous TFE direct integration approach was developed for nonlinear structural-dynamics problems by Lee and Kwak [22]. In 1996, Fung proposed higher-order L-stable direct integration TFE methods based on Hermite polynomials [23].

Methods that can be proven unconditionally stable for linear problems may lead to instability when applied to nonlinear dynamical problems. Therefore, designing numerical

strategies for nonlinear dynamics is particularly challenging. The TFE approach as a general class of integrators for nonlinear and chaotic problems was established in 2016 by Jog et al. [24]. The study further revealed that a majority of existing methods recognized for their robustness in addressing periodic, parabolic or hyperbolic problems can be formulated within the framework of time finite elements. This approach, therefore, serves as a unifying foundation for time-stepping techniques employed across various fields.

In more recent years, the potential of TFE based integrators has again rejuvenated the attention of researchers from various fields. Antonietti et al. [25, 26] have developed discontinuous Galerkin spectral element methods for simulating seismic-wave propagation and showed that the resulting discrete formulations are well-posed, stable, and retain a super-optimal rate of convergence with respect to the step-size and polynomial order. Tamma et al. [27] have presented Petrov-Galerkin based unified framework to design high-order algorithms. This approach yields various unconditionally stable (A/L-stable) algorithms with controllable numerical dissipation. However, the applications have been limited to linear elastodynamics problems.

Furthermore, Bauchau [28] has recently employed Hamilton's two-field principle to develop time-continuous and time-discontinuous TFE schemes for constrained multibody systems. However, the computer-implementation has been demonstrated in problem-specific minimal coordinates, instead of more generic matrix formulations. In context of direct integration methods, Kim and Reddy [29] have proposed a TFE based direct integration method that not only gives higher order accuracy for linear as well as nonlinear problems, but also gives active numerical damping control using a single parameter. This scheme is based on modified-weighted residual method where a single weighting function can be used to generate linearly-independent relations at all the nodes in the finite element system. However, the method has been employed on structural-dynamics ordinary differential equations (ODEs). When dealing with DAEs, order reduction with ODE integrators is a common phenomena [30]. Hence, it is important to analyze the efficacy of TFE integrators in context of DAEs.

1.2 Contributions of this study

The focus of this work is the implementation of higher order integration schemes within S^3 Lie-group DAE framework, while accounting friction in joints. Hence, the discussion on Lie-theory is bypassed. The literature on Lie-group theory in context of dynamics is rich and readers may refer to primary contributions such as [10, 11, 13, 14, 31–33] for more details. To add context to the presented report, the efficacy of the approach [1–3] with its reported computational advantages was acknowledged. However, two primary limitations of this approach were considered: (1) Evaluation of Jacobian-equivalent matrices for friction/arbitrary nonlinear loads will require case-by-case transformations, as provided for kinematic constraints in [1–3]; (2) First order integration with the use of computed Jacobian-equivalent matrices. Aiming at these two primary deliverables, the key contributions of the reported work are as follows:

1. Since simulating friction is essential to multibody systems, and intricacies that arise with nonlinear loads need to be addressed to complete the Lie-group framework. The first contribution of the submitted manuscript provides a generic derivative-based definition of Jacobian-equivalent matrices for arbitrary nonlinear loads, including friction. This contribution to include friction in Lie-group based frameworks is novel and, to the best of our knowledge, has not been addressed in existing Lie-group frameworks.
2. Higher order direct numerical integration methods for index-3 DAEs are implemented in Lie-Group framework. For second order accuracy, implementation of Newmark- β , method is presented.

3. A time-finite element (TFE) family of integrators for DAEs is formulated, targeting third and fifth order accuracy. The performance of the discussed numerical methods is demonstrated and compared using numerical experiments, focusing on the friction-induced effects.
4. The performance of time-finite element integrators has not been studied considering friction-induced nonlinearity to the best of our knowledge. The presented analysis essentially aims to take a holistic approach of assessing these schemes in a Lie-group framework equipped with friction modeling capability.

To the best of our knowledge, the above contributions are novel and have not been reported in existing literature.

The remainder of this paper is structured as follows. Section 2 outlines the absolute-coordinate index-3 DAE formulation with quaternions as generalized rotation coordinates and utilizes their Lie-group properties to circumvent the additional normalization constraint. Section 3 introduces a TFE based direct integration applied with S^3 group on index-3 DAEs. A general methodology for nonlinear DAEs with friction is proposed and the formulation of corresponding Newton-iteration is described in detail. In addition, the Sect. 3.3 demonstrates the implementation of time-tested Newmark- β and trapezoidal integration schemes in S^3 group formulation. Section 4 presents the numerical experiments on a rigid-pendulum case study. The performance of numerical integrators is assessed on two cases: frictionless pendulum and with friction in account. Section 5 concludes with key observations and a discussion on further scope of the work.

2 Lie group formulation of DAE of multibody systems with friction

The index-3 DAE describing a constrained multibody system dynamics in absolute coordinates ($\mathbf{r}$ - $\mathbf{A}$ formulation) is as follows:

$$\mathbf{M}\ddot{\mathbf{r}} + \Phi_{\mathbf{r}}^T \boldsymbol{\lambda} = \mathbf{f}(\mathbf{r}, \mathbf{A}), \quad (1a)$$

$$\mathbf{J}\dot{\boldsymbol{\omega}}' + \tilde{\boldsymbol{\omega}}' \mathbf{J} \boldsymbol{\omega}' + \Pi(\Phi)^T \boldsymbol{\lambda} = \mathbf{n}'(\mathbf{r}, \mathbf{A}), \quad (1b)$$

$$\Phi(\mathbf{r}, \mathbf{A}) = \mathbf{0}. \quad (1c)$$

Here, $\mathbf{M} \in \mathbb{R}^{3N \times 3N}$ and $\mathbf{J} \in \mathbb{R}^{3N \times 3N}$ are the global mass and inertia matrices, respectively, of the complete system comprising of N rigid bodies. For each body i , the Cartesian coordinates $\mathbf{r}_i \in \mathbb{R}^3$ denote the spatial position of the respective body, and the rotation matrix $\mathbf{A}_i \in \mathbb{R}^{3 \times 3}$ represents its orientation with respect to global reference frame. The acting loads $\mathbf{f}$ and torques $\mathbf{n}'$ are computed as functions of combined vector of spatial positions $\mathbf{r} = [\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_N^T]^T$, and orientations $\mathbf{A} = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_N \mid \mathbf{A}_i \in \text{SO}(3)\}$. With algebraic constraints defined by the set of equations $\Phi(\mathbf{r}, \mathbf{A}) = \mathbf{0} \in \mathbb{R}^C$, the Jacobian matrix $\Phi_{\mathbf{r}} \in \mathbb{R}^{C \times 3N}$ represents partial derivatives with respect to Cartesian coordinates.

The rotation matrix of i th body $\mathbf{A}_i$ is computed as a matrix-valued function of either Euler-parameters $\mathbf{p}_i \in S^3$, $\mathbf{p}_i = [e_0, \mathbf{e}^T]^T$ in $\mathbf{r}$ - $\mathbf{p}$ formulation, or of Euler-angles $\boldsymbol{\epsilon}_i \in \mathbb{R}^3$, $\boldsymbol{\epsilon}_i = [\phi, \theta, \psi]^T$ in $\mathbf{r}$ - $\boldsymbol{\epsilon}$ formulation [2, 3, 34]. The state variables in (1a)–(1c) comprise of linear and angular accelerations $\ddot{\mathbf{r}} \in \mathbb{R}^{3N}$ and $\dot{\boldsymbol{\omega}}' \in \mathbb{R}^{3N}$, respectively, and the Lagrange multipliers $\boldsymbol{\lambda} \in \mathbb{R}^C$ associated with the constraint manifold $\Phi \in \mathbb{R}^C$. Furthermore, the entities $\mathbf{f}$ and $\mathbf{n}'$ represent the externally acting forces and torques on the bodies, respectively, that are often nonlinear functions of the position and orientation variables. The operator

“ $\cdot$ ” produces the skew-symmetric matrix form of the vector over which it operates, for instance, $\tilde{\omega}'$ in equation (1b) converts the body-fixed vector of angular velocity ω' into a skew-symmetric matrix.

Since the rotation matrix is always associated with a body-fixed vector s' (represented globally as $s = As'$), the operator Π essentially reflects how the perturbation in A changes the orientation of s . As defined by Kissel et al. in [1–3], this infinitesimal change is computed as $\delta s = \Pi(As')\delta\theta = -A\tilde{s}'\delta\theta$, where θ represents infinitesimal local-rotation. Similarly, the coefficient matrix $\Pi(\Phi) \in \mathbb{R}^{C \times 3N}$ maps the infinitesimal first-order perturbation in orientation to the first-order perturbation in constraints Φ . In a Newton-Raphson scheme, these corrections in local orientation θ , $\tilde{\theta} \in \mathfrak{so}(3)$, are then used to update the rotation matrix A at every iteration step, as delineated by Kissel et al. in [1–3]. Here, $\mathfrak{so}(3)$ is the Lie-Algebra of the $SO(3)$ manifold parametrized by local rotations θ .

The solution strategy for solving the frictionless DAE (1a)–(1c), as established by Kissel et al. [1, 3], performs first-order implicit numerical integration, and lays the groundwork for developing higher-order integration schemes. Like most implicit integration schemes, this strategy also requires solving nonlinear equations of the following form:

$$\mathbf{d}(\mathbf{r}, \mathbf{A}, \boldsymbol{\lambda}) = \mathbf{0} \in \mathbb{R}^{6N+C}, \quad \text{where } \mathbf{A} = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_N \mid \mathbf{A}_i \in SO(3)\}. \quad (2)$$

This is achieved by running Newton-Raphson iterations at each time-step, which requires k iterations each solving a linear equation $\tilde{\mathbf{D}}^{k-1} \delta^k = -\mathbf{d}(\mathbf{r}^{k-1}, \mathbf{A}^{k-1}, \boldsymbol{\lambda}^{k-1})$, where, $\delta = [\delta \mathbf{r}^T, \delta \theta^T, \delta \boldsymbol{\lambda}^T]^T$ is the vector containing infinitesimal corrections in position $\mathbf{r} \in \mathbb{R}^{3N}$, body-frame Euler-rotation vector $\theta \in \mathbb{R}^{3N}$, and Lagrange multipliers $\boldsymbol{\lambda} \in \mathbb{R}^C$. The Euler-rotation vector θ essentially contains the components of angle rotated by the body-fixed coordinate-frame about body-fixed axis directed along angular velocity vector ω' , within a specified time-interval $[t_0, t]$, such that [14]:

$$\tilde{\theta}(t) = \text{dexp}_{\tilde{\theta}}^{-1}(\tilde{\omega}'), \quad \theta(t_0) = \mathbf{0}, \quad (3)$$

where $\text{dexp}_{\tilde{\theta}}^{-1}$ is the inverse of the differential of the exponent map. This nonlinear map is essential to take into account because large rotations are defined in $SO(3)$, which means that integration of (3) yields a orientation-update iteration scheme in terms of angular velocity field ω' . Furthermore, in formulations based on Euler angles or quaternions, the rotation matrix A can be parameterized by either of the arguments, i.e., quaternions or Euler-angles. However, the true space of 3-D rotations is the $SO(3)$ manifold while the arguments are only a coordinate representation of $SO(3)$ space. Therefore, systems involving large-rotational motions must evolve on $SO(3)$. After k iterations, the solution converges to numerical precision ε such that $\|\mathbf{d}^k\| \leq \varepsilon$. For instance, the k th Newton-Raphson iteration on constraint equation (1c) yields:

$$\Phi(\mathbf{r}, \mathbf{A}) = \mathbf{0} \implies \Phi_{\mathbf{r}}^{k-1} \delta \mathbf{r}^k + \Pi(\Phi^{k-1}) \delta \theta^k = -\Phi(\mathbf{r}^{k-1}, \mathbf{A}^{k-1}). \quad (4)$$

Here, the sub-Jacobians $\Phi_{\mathbf{r}}$ corresponding to corrections $\delta \mathbf{r}$ are simply the partial derivatives of constraint manifold Φ , with respect to $\mathbf{r}$. However, computing the coefficient matrix $\Pi(\Phi)$ corresponding to infinitesimal local rotations $\delta \theta$ is a relatively more complex process. As defined by Kissel et al. in [3], it requires evaluation of time-derivatives of the entity of interest and representing them as matrix-vector multiplications of a coefficient matrix and angular-velocity ω' . Kissel et al. describe this process in detail in [2], and give explicit analytically derived elements of $\Pi(\Phi)$ for all fundamental kinematic constraints.

2.1 Modeling friction in Lie-group DAE formulation

Consider the set of all the points in time $\tilde{\mathcal{T}} = \{t \in \mathbb{R} \mid t \geq 0\}$. Let the k th finite time interval be defined as $\mathcal{T}^k = \{\mathcal{T}^k \subset \tilde{\mathcal{T}} \mid t \in \mathcal{T}^k \implies t_1^k \leq t \leq t_2^k, t_1^k = \min(\mathcal{T}^k), t_2^k = \max(\mathcal{T}^k)\}$, such that $\tilde{\mathcal{T}} = \bigcup_{k=1}^n \mathcal{T}^k$. The inverse of the differential of the exponent map (3) given by Terze et al. [14] is essentially an ordinary differential equation with initial conditions such that at every time-step, the starting point is the current pose itself. Hence emerges the initial condition $\theta(t_{init}) = \mathbf{0}$, where $t_{init} \in \mathcal{T}_{init} := \{t \in \tilde{\mathcal{T}} \mid t \in \bigcup_{k=1}^n \min(\mathcal{T}^k)\}$. Since, the equations (3) hold true for all time intervals $\mathcal{T}^k \subseteq \tilde{\mathcal{T}}$, they can be coupled with the dynamics DAEs for each time interval $\mathcal{T}^k$, such that the initial conditions for the pose hold true at the beginning of each interval. Recent work [5] extends the DAE (1a)–(1c) to include friction in joints and yields the following modified DAE,

$$\mathbf{M}\ddot{\mathbf{r}} + \Phi_r^T \lambda = \mathbf{f}(\mathbf{r}, \mathbf{A}) + \mathbf{f}_{\text{fric}}(\mathbf{r}, \dot{\mathbf{r}}, \omega', \mathbf{A}, \lambda), \quad (5a)$$

$$\mathbf{J}\dot{\omega}' + \tilde{\omega}' \mathbf{J}\omega' + \Pi(\Phi)^T \lambda = \mathbf{n}'(\mathbf{r}, \mathbf{A}) + \mathbf{n}'_{\text{fric}}(\mathbf{r}, \dot{\mathbf{r}}, \omega', \mathbf{A}, \lambda), \quad (5b)$$

$$\Phi(\mathbf{r}, \mathbf{A}) = \mathbf{0}, \quad (5c)$$

where $\mathbf{f}_{\text{fric}}$ and $\mathbf{n}'_{\text{fric}}$ represent the friction induced forces and torques, respectively.

Furthermore, the inverse of the differential of the exponent map (3) can be expressed in the following two forms: as a series of commutators (6a), as well as a closed form (6b) [14]:

$$\text{dexp}_{-\tilde{\theta}}^{-1}(\tilde{\omega}') = \tilde{\omega}' + \frac{1}{2}[\tilde{\theta}, \tilde{\omega}'] + \frac{1}{12}[\tilde{\theta}, [\tilde{\theta}, \tilde{\omega}']] + \cdots = \sum_{j=0}^{\infty} \frac{B_j}{j!} (-\text{ad}_{\tilde{\theta}}^j(\tilde{\omega}')), \quad (6a)$$

$$\text{dexp}_{-\tilde{\theta}}^{-1}(\tilde{\omega}') = \tilde{\omega}' + \frac{1}{2}\tilde{\theta}\tilde{\omega}' + \frac{\|\tilde{\theta}\| \cot \frac{\|\tilde{\theta}\|}{2} - 2}{2\|\tilde{\theta}\|^2} \tilde{\theta}^2 \tilde{\omega}', \quad (6b)$$

where, B_j are the Bernoulli numbers and $\text{ad}_{\tilde{\theta}}^j$ is the adjoint operator. In [14], the authors suggest truncating the series (6a) at p^{th} index, where for the required k th order of accuracy, $p \geq k - 2$. This powerful property implies that upto fifth order of accuracy could be potentially achieved with just first three terms. It is hence of academic interest to assess the effectiveness of truncation index $p = 3$ for order preservation, in an implicit integration framework, i.e., with large time-step sizes. Moreover, (6a) preserves the skew-symmetric structure of the output, that makes one-to-one conversion of (6a) into vector space straight-forward. The first three terms of (6a), on conversion to vector-space, yield following ODE:

$$\omega' = \mathbf{H}(\theta)\dot{\theta}, \quad \mathbf{H}(\theta) = \begin{bmatrix} \mathbf{I}_1^T \\ \mathbf{I}_2^T \\ \mathbf{I}_3^T \end{bmatrix}, \quad \forall t \in \tilde{\mathcal{T}}, \quad (7)$$

$$\theta(t_i) = \mathbf{0}, \quad \forall t_i \in \mathcal{T}_{init}$$

where, $\mathbf{H}(\theta)$ is computed as shown in Appendix A, and $\theta \in \mathfrak{so}(3)$. Differentiating (7) yields the following relationship with acceleration:

$$\dot{\omega}' = \mathbf{H}(\theta)\ddot{\theta} + \mathbf{W}(\theta, \dot{\theta})\dot{\theta}, \quad \mathbf{W}(\theta, \eta) = (\mathbf{H}(\theta)\eta)_{\theta} \quad (8)$$

Just like $\mathbf{H}(\theta)$, the matrix $\mathbf{W}(\theta, \eta)$ can also be evaluated symbolically. Substituting equations (7) and (8) into (5a)–(5c), the DAEs can be transformed into absolute coordinates form where $\mathbf{u} = [\mathbf{r}^T, \theta^T]^T$ are absolute coordinates. The resulting DAEs for rotational motion (5b), for each time-interval $\mathcal{T}^k \subseteq \bar{\mathcal{T}}$ can be re-written as:

$$\hat{\mathbf{J}}(\theta)\ddot{\theta} + \hat{\mathbf{C}}(\theta, \dot{\theta})\dot{\theta} + \Pi(\Phi)^T \lambda = \mathbf{n}'(\mathbf{r}, \mathbf{A}) + \mathbf{n}'_{\text{fric}}(\mathbf{r}, \dot{\mathbf{r}}, \omega', \mathbf{A}, \lambda), \quad \theta(t_0) = \mathbf{0}, \quad (9a)$$

$$\hat{\mathbf{J}}(\theta) = \mathbf{J}\mathbf{H}(\theta), \quad \hat{\mathbf{C}}(\theta, \dot{\theta}) = \mathbf{J}\mathbf{W}(\theta, \dot{\theta}) + \widetilde{(\mathbf{H}(\theta)\dot{\theta})} \hat{\mathbf{J}}(\theta), \quad t_0 = \min(\mathcal{T}^k). \quad (9b)$$

Now, the DAE (5a)–(5c) can be modified such that the rotation matrix $\mathbf{A}$ is represented as a matrix-valued function of unit-quaternions, i.e., $\mathbf{A} = \psi(\mathbf{p}) \iff \mathbf{p}^T \mathbf{p} = 1$ where $\psi: \mathbb{R}^4 \rightarrow \mathbb{R}^{3 \times 3}$. The modified DAE (5a)–(5c) can thus be written in fully-implicit form as follows:

$$\mathbf{d}(\mathbf{q}, \mathbf{v}, \lambda, \dot{\mathbf{v}}) = \begin{bmatrix} \mathbf{d}_1 \\ \mathbf{d}_2 \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{M}}\dot{\mathbf{v}} + \mathbf{F}_A(\mathbf{q}, \mathbf{v}_a, \lambda) + \mathbf{F}_{rot}(\theta, \dot{\theta}) \\ \Phi(\mathbf{q}) \end{bmatrix} = \mathbf{0}, \quad (10a)$$

$$\bar{\mathbf{M}} = \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{J}}(\theta) \end{bmatrix}, \quad \dot{\mathbf{v}} = \begin{bmatrix} \dot{\mathbf{r}} \\ \dot{\theta} \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} \mathbf{r} \\ \theta \end{bmatrix}, \quad \mathbf{v}_a = \begin{bmatrix} \dot{\mathbf{r}} \\ \omega' \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} \mathbf{r} \\ \mathbf{p} \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} \mathbf{r} \\ \theta \end{bmatrix},$$

$$\mathbf{F}_A(\mathbf{q}, \mathbf{v}_a, \lambda) = \begin{bmatrix} \Phi_r^T(\mathbf{q})\lambda - \mathbf{f}(\mathbf{q}) - \mathbf{f}_{\text{fric}}(\mathbf{q}, \mathbf{v}_a, \lambda) \\ \Pi^T(\Phi(\mathbf{q}))\lambda - \mathbf{n}'(\mathbf{q}) - \mathbf{n}'_{\text{fric}}(\mathbf{q}, \mathbf{v}_a, \lambda) \end{bmatrix}, \quad (10b)$$

$$\mathbf{F}_{rot}(\theta, \dot{\theta}) = \begin{bmatrix} \mathbf{0} \\ \hat{\mathbf{C}}(\theta, \dot{\theta})\dot{\theta} \end{bmatrix},$$

$$\mathbf{v}_a = \hat{\mathbf{H}}(\theta)\mathbf{v}, \quad \hat{\mathbf{H}}(\theta) = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{H}(\theta) \end{bmatrix}.$$

The sub-Jacobians of general nonlinear forcing functions, such as friction forces and torques in DAE (5a)–(5c), corresponding to infinitesimal corrections $\delta \mathbf{r}, \delta \dot{\mathbf{r}}, \delta \ddot{\mathbf{r}}, \delta \lambda, \delta \omega', \delta \dot{\omega}'$ can be simply obtained as their partial derivatives with respect to $\mathbf{r}, \dot{\mathbf{r}}, \ddot{\mathbf{r}}, \lambda, \omega', \dot{\omega}'$ respectively. This can be achieved via a gamut of numerical and/or auto-differentiation tools. However, computing the coefficient matrix for nonlinear (friction) forces and torques corresponding to infinitesimal local rotation $\delta \theta$ can only be evaluated on a case-by-case basis. This limits the applicability of $\mathbf{r}$ – $\mathbf{A}$ formalism when arbitrary complex nonlinear forcing functions are included. In contrast to $\mathbf{r}$ – $\mathbf{A}$ formulation, $\mathbf{r}$ – $\mathbf{p}$ and $\mathbf{r}$ – ϵ formulations are devoid of this limitation as the equivalent coefficient matrices are simply computed as Jacobians with respect to $\mathbf{p}$ and ϵ respectively. Hence, a general representation of these coefficient matrices in absolute coordinates is an open problem. Because of wide availability of numerical and automatic differentiation toolboxes, it is preferable to have the general representation as a composition of derivatives with respect to the global orientation parameters $\mathbf{p}$ or ϵ .

2.2 General representation of coefficient matrix of local-rotations using Lie-group properties of quaternions

Computation of derivatives with respect to Euler-parameters (unit-quaternions) $\mathbf{p}$ allows to circumvent the issue of gimbal lock. The mapping between the body-fixed angular velocity vector ω' and the time derivative of $\mathbf{p}$ is given using the orthogonal matrix $\mathbf{G}(\mathbf{p})$ as follows

[4]:

$$\mathbf{G}(\mathbf{p}) := [-\mathbf{e} - \tilde{\mathbf{e}} + e_0 \mathbf{I}_3],$$

$$\dot{\mathbf{p}} = \frac{1}{2} \mathbf{G}(\mathbf{p})^\top \boldsymbol{\omega}' \implies \dot{\mathbf{p}} = \frac{1}{2} \mathcal{G}(\boldsymbol{\omega}') \mathbf{p}, \quad \text{where } \mathcal{G} = \begin{bmatrix} 0 & -\boldsymbol{\omega}'^\top \\ \boldsymbol{\omega}' & -\tilde{\boldsymbol{\omega}}' \end{bmatrix}. \quad (11)$$

The bilinear mapping, from [4], proposes that the matrix $\mathcal{G}(\boldsymbol{\omega}')$ belongs to the Lie-algebra of quaternions. Further, in finite-element based formalism, the velocities and positions are approximated using polynomial bases as time-dependent functions [22, 27, 29]. Following this, the angular-velocity signals $\boldsymbol{\omega}'(t)$ and orientation parameters $\mathbf{p}(t)$ are treated as time-dependent functions represented by polynomial bases. Hence, the resulting first order system (11) at a time t is as follows:

$$\dot{\mathbf{p}}(t) = \frac{1}{2} \mathcal{G}(\boldsymbol{\omega}'(t)) \mathbf{p}(t). \quad (12)$$

Since the above system is non-autonomous, the general solution is represented as follows, where $\mathbf{p}_0$ is the initial value at $t = t_0$ and $\boldsymbol{\theta}(t_0) = \mathbf{0}$:

$$\mathbf{p}(t) = \exp\left(\int_{t_0}^t \frac{1}{2} \mathcal{G}(\boldsymbol{\omega}'(t)) dt\right) \mathbf{p}_0 \implies \mathbf{p}(t) = \exp\left(\frac{1}{2} \mathcal{G}(\boldsymbol{\theta}(t))\right) \mathbf{p}_0, \quad (13)$$

and after algebraic manipulations, such as in [4, 5], an exact representation of the matrix exponential is as follows:

$$\exp\left(\frac{1}{2} \mathcal{G}(\boldsymbol{\theta}(t))\right) = \cos\left(\frac{\|\boldsymbol{\theta}(t)\|}{2}\right) \mathbf{I}_4 + \frac{\sin\left(\frac{\|\boldsymbol{\theta}(t)\|}{2}\right)}{\|\boldsymbol{\theta}(t)\|} \mathcal{G}(\boldsymbol{\theta}(t)). \quad (14)$$

This exponential integration scheme on unit-quaternions automatically satisfies the normalization constraint $\mathbf{p}^\top \mathbf{p} = 1$ [4]. The operator $\mathcal{G}$ applied to any input argument yields a skew-symmetric matrix containing the elements of the argument vector, as illustrated in equation (11). Moreover, unit quaternions are not a complete solution for large rotations, but just a chart. In true sense, quaternion Lie group is double cover of $\text{SO}(3)$, which implies that two quaternions describe the same rotation [11, 14]. When the argument is the Euler rotation vector $\boldsymbol{\theta}$, the resulting matrix $\mathcal{G}(\boldsymbol{\theta})$ is simply composed of elements of vector $\boldsymbol{\theta}$ and hence is linear in $\boldsymbol{\theta}$. Therefore, the discrete equation at n th time-step $[t_{n-1}, t_n]$ equivalent to continuous form (13), can be represented as:

$$\mathbf{p}_n = \exp\left(\frac{1}{2} \mathcal{G}(\boldsymbol{\theta}_n)\right) \mathbf{p}_{n-1} = \exp\left(\mathcal{G}\left(\frac{\boldsymbol{\theta}_n}{2}\right)\right) \mathbf{p}_{n-1}. \quad (15)$$

In a Newton-Raphson iteration, evolution of $\mathbf{p}_n$ at k th iteration from the previous time-step $\mathbf{p}_{n-1}$ takes the form $\mathbf{p}_n^k = \exp\left(\mathcal{G}\left(\frac{\boldsymbol{\theta}^k}{2}\right)\right) \mathbf{p}_{n-1}$. In the following analysis, the subscript denoting n th time-step has been dropped for brevity. For small perturbation in $\boldsymbol{\theta}^k$, the matrix exponential can be approximated with first-order expansion: $\exp\left(\mathcal{G}\left(\frac{\delta \boldsymbol{\theta}^k}{2}\right)\right) \approx \mathbf{I}_4 + \mathcal{G}\left(\frac{\delta \boldsymbol{\theta}^k}{2}\right) + O(\delta \boldsymbol{\theta}^{k^2})$. Subsequently, a perturbation $\delta \boldsymbol{\theta}^k$ in body-frame rotation produces

the following correction in respective $\mathbf{p}^k$:

$$\begin{aligned}
 \mathbf{p}^k - \mathbf{p}^{k-1} &= \exp\left(\mathcal{G}\left(\frac{\boldsymbol{\theta}^{k-1} + \delta\boldsymbol{\theta}^k}{2}\right)\right) \mathbf{p}_{n-1} - \exp\left(\mathcal{G}\left(\frac{\boldsymbol{\theta}^{k-1}}{2}\right)\right) \mathbf{p}_{n-1}, \\
 \Rightarrow \delta\mathbf{p}^k &= \exp\left(\mathcal{G}\left(\frac{\boldsymbol{\theta}^{k-1}}{2}\right)\right) \left(\exp\left(\mathcal{G}\left(\frac{\delta\boldsymbol{\theta}^k}{2}\right)\right) - \mathbf{I}_4\right) \mathbf{p}_{n-1}, \\
 \Rightarrow \delta\mathbf{p}^k &\approx \exp\left(\mathcal{G}\left(\frac{\boldsymbol{\theta}^{k-1}}{2}\right)\right) \left(\mathcal{G}\left(\frac{\delta\boldsymbol{\theta}^k}{2}\right)\right) \mathbf{p}_{n-1}, \\
 \Rightarrow \delta\mathbf{p}^k &\approx \left[\frac{1}{2} \exp\left(\mathcal{G}\left(\frac{\boldsymbol{\theta}^{k-1}}{2}\right)\right) \mathbf{G}(\mathbf{p}_{n-1})\right] \delta\boldsymbol{\theta}^k =: \mathbf{T}_{\text{exp}}(\boldsymbol{\theta}^{k-1}) \delta\boldsymbol{\theta}^k.
 \end{aligned} \tag{16}$$

Therefore, the matrix $\mathbf{T}_{\text{exp}}(\boldsymbol{\theta}^{k-1})$ can be used to compute change in global orientation $\delta\mathbf{p}$ for infinitesimal corrections in local-frame rotation $\boldsymbol{\theta}$. This means, that infinitesimal change in any arbitrary nonlinear vector-valued function of generalized coordinates and Euler rotations, $\mathbf{q} = [\mathbf{r}^T \mathbf{p}^T]^T$ and $\delta\boldsymbol{\theta}$ respectively, containing p elements, $\mathbf{g}(\mathbf{q}, \boldsymbol{\theta}) : \mathbb{R}^6 \times S^3 \rightarrow \mathbb{R}^p$, can be computed by using $\mathbf{T}_{\text{exp}}$ from equations (16). Hence, the coefficient matrix corresponding to local rotations $\Pi(\mathbf{g})$ assumes a general form given as follows:

$$\begin{aligned}
 \delta\mathbf{g}^k &= \mathbf{g}_{\mathbf{q}}(\mathbf{q}^{k-1}, \boldsymbol{\theta}^{k-1}) \delta\mathbf{q}^k + \mathbf{g}_{\boldsymbol{\theta}}(\mathbf{q}^{k-1}, \boldsymbol{\theta}^{k-1}) \delta\boldsymbol{\theta}^k \\
 \Rightarrow \delta\mathbf{g}^k &= \mathbf{g}_{\mathbf{r}}(\mathbf{q}^{k-1}, \boldsymbol{\theta}^{k-1}) \delta\mathbf{r}^k + [\mathbf{g}_{\mathbf{p}}(\mathbf{q}^{k-1}, \boldsymbol{\theta}^{k-1}) \mathbf{T}_{\text{exp}}(\boldsymbol{\theta}^{k-1}) + \mathbf{g}_{\boldsymbol{\theta}}(\mathbf{q}^{k-1}, \boldsymbol{\theta}^{k-1})] \delta\boldsymbol{\theta}^k, \\
 \Rightarrow \delta\mathbf{g}^k &= [\mathbf{g}_{\mathbf{r}}^{k-1} \quad \Pi(\mathbf{g}^{k-1})] \begin{bmatrix} \delta\mathbf{r}^k \\ \delta\boldsymbol{\theta}^k \end{bmatrix}.
 \end{aligned} \tag{17}$$

In equations (17), the action of right multiplication of Jacobian $\Phi_{\mathbf{p}}$ with matrix $\mathbf{T}_{\text{exp}}$ essentially transports $\Phi_{\mathbf{p}}$ to an orientation rotated by $\boldsymbol{\theta}$. Hence, $\mathbf{T}_{\text{exp}}$ can be defined as a transportation matrix on S^3 group. Furthermore, we define a set of spatial coordinates $\mathbf{u} = [\mathbf{r}^T, \boldsymbol{\theta}^T]^T$, corresponding to any concerned body at each time-step, that represent: (1) Cartesian coordinates in three-dimensional space, and (2) the angular rotation of the body-frame about a body-fixed axis, in a specified time-interval. Therefore, a generalized transformation operator Π can be defined as follows.

Definition 2.1 Consider the set $\mathcal{B} = \{\mathbf{a}(\mathbf{p}, \boldsymbol{\theta}) \mid \forall \mathbf{p} \in S^3, \tilde{\boldsymbol{\theta}} \in \mathfrak{so}(3)\}$, the image of S^3 through a vector valued function $\mathbf{a}(\mathbf{p}, \boldsymbol{\theta}) : S^3 \times \mathfrak{so}(3) \rightarrow \mathbb{R}^m$. The operator $\Pi : \mathcal{B} \rightarrow \mathbb{R}^{m \times 3}$ is a matrix valued function such that its action on an arbitrary vector $\mathbf{a}(\mathbf{p}, \boldsymbol{\theta}) \in \mathcal{B}$, is obtained by right-multiplication of the Jacobian $\mathbf{a}_{\mathbf{p}}$ with the transportation matrix $\mathbf{T}_{\text{exp}}$ parameterized by the Euler rotation $\boldsymbol{\theta}$, i.e.,

$$\Pi(\mathbf{a}) = \mathbf{a}_{\mathbf{p}} \mathbf{T}_{\text{exp}}(\boldsymbol{\theta}) + \mathbf{a}_{\boldsymbol{\theta}}. \tag{18}$$

The action of operator Π yields a coefficient matrix that maps the variations in $\mathbf{a}$ with respect to unit-quaternions $\mathbf{p}$ to the corresponding variations with respect to the Euler rotation $\boldsymbol{\theta}$.

Furthermore, the coefficient matrix $\Pi(\Phi)$ corresponding to constraint torques can be either computed from analytical expressions in [1–3], or it can be equivalently computed as $\Pi(\Phi) \equiv \Phi_{\mathbf{p}} \mathbf{T}_{\text{exp}}$. To demonstrate this principle experimentally, let $\Pi^*(\Phi)$ signify the analytically evaluated matrix and $\Phi_{\mathbf{p}}$ can be evaluated either analytically [6, 34] or numerically. For worst case scenario, it is vital that the error dynamics be reported with $\Phi_{\mathbf{p}}$ evaluated numerically; for instance, using a central finite-difference scheme. As a measure of error,

the value of Frobenius error norm η can be stored at each simulation time-step, and can be computed using equation (19):

$$\eta = \|\Pi^*(\Phi) - \Phi_p \mathbf{T}_{\text{exp}}\|_F. \quad (19)$$

Using Definition 2.1, a generalized transformation operator can be defined for the complete set of generalized coordinates.

Definition 2.2 Consider the set $\mathcal{F} = \{\mathbf{g}(\mathbf{q}, \boldsymbol{\theta}) \mid \forall \mathbf{q} \in \mathbb{R}^3 \times S^3, \tilde{\boldsymbol{\theta}} \in \mathfrak{so}(3)\}$. The operator $\Theta : \mathcal{F} \rightarrow \mathbb{R}^{6 \times 6}$ is a matrix valued function whose action on an arbitrary vector $\mathbf{g}(\mathbf{q}, \boldsymbol{\theta}) \in \mathcal{F}$, yields a coefficient matrix that maps the variations in $\mathbf{g}$ with respect to generalized coordinates $\mathbf{q}$ to the corresponding variations with respect to new set of defined coordinates $\mathbf{u}$, i.e.,

$$\Theta(\mathbf{g}) = [\mathbf{g}_r \quad \mathbf{g}_p \mathbf{T}_{\text{exp}} + \mathbf{g}_\theta] = [\mathbf{g}_r \quad \Pi(\mathbf{g})]. \quad (20)$$

The operator Θ acts on the arbitrary vector $\mathbf{g}(\mathbf{q}, \boldsymbol{\theta})$ and transforms into the coefficient matrix using equation (20). Using the operator Θ from Definition 2.2 allows to rewrite equation (17) as follows:

$$\begin{aligned} \delta \mathbf{g}^k &= [\mathbf{g}_r^{k-1} \quad \Pi(\mathbf{g}^{k-1})] \begin{bmatrix} \delta \mathbf{r}^k \\ \delta \boldsymbol{\theta}^k \end{bmatrix}, \\ \implies \delta \mathbf{g}^k &= \mathbf{g}_q^{k-1} \delta \mathbf{q}^k = \Theta(\mathbf{g}^{k-1}) \delta \mathbf{u}^k. \end{aligned} \quad (21)$$

The friction loads (forces and torques) for fundamental joints are evaluated in the reference [5]. It is clear that the friction loads are essentially intricate functions of $\mathbf{q}$, $\mathbf{v}$, $\boldsymbol{\lambda}$. While the Jacobians of friction loads with respect to $\mathbf{v}$ and $\boldsymbol{\lambda}$ can be directly computed as partial derivatives, it is important to convert Jacobian with respect to $\mathbf{q}$ to the absolute coordinate counterpart. This is achieved by the proposed operator $\mathbf{T}_{\text{exp}}$ and the approach is general in the sense as the conversion does not depend on the type of joint.

3 Higher order time integration in absolute coordinates using TFE

For a numerical solution, one starts with the known states $\mathbf{q}_{n-1}$, $\boldsymbol{\theta}_{n-1}$, $\mathbf{v}_{n-1}$, $\boldsymbol{\lambda}_{n-1}$, $\dot{\mathbf{v}}_{n-1}$ at t_{n-1} and computes the unknown states $\mathbf{q}_n$, $\boldsymbol{\theta}_n$, $\mathbf{v}_n$, $\boldsymbol{\lambda}_n$, $\dot{\mathbf{v}}_n$ at t_n . The solution of the implicit DAE (10a)–(10b) requires a Newton-Raphson iteration, starting with some initial values. Using the definition of the operator Θ from equations (20), the k th Newton-Raphson iteration for solving equation (10a) can be stated as follows:

$$\begin{aligned} &\left[\bar{\mathbf{M}}^{k-1} \delta \dot{\mathbf{v}}_n^k + \left((\bar{\mathbf{M}} \dot{\mathbf{v}})_u^{k-1} + \mathbf{F}_{rot \dot{\mathbf{u}}}^{k-1} \right) \delta \mathbf{u}_n^k + \mathbf{F}_{rot \dot{\mathbf{u}}}^{k-1} \delta \mathbf{v}_n^k + \mathbf{F}_A^{k-1} \delta \mathbf{v}_{a_n}^k + \mathbf{F}_{\lambda}^{k-1} \delta \boldsymbol{\lambda}_n^k + \Theta(\mathbf{F}_A^{k-1}) \delta \mathbf{u}_n^k \right] \\ &\quad \Theta(\Phi^{k-1}) \delta \mathbf{u}_n^k \\ &= -\mathbf{d}(\mathbf{q}_n^{k-1}, \mathbf{v}_{a_n}^{k-1}, \boldsymbol{\lambda}_n^{k-1}, \dot{\mathbf{v}}_n^{k-1}), \quad (\bar{\mathbf{M}} \dot{\mathbf{v}})_u = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{J} \mathbf{W}(\boldsymbol{\theta}, \ddot{\boldsymbol{\theta}}) \end{bmatrix}, \end{aligned} \quad (22a)$$

$$\begin{aligned} \mathbf{u}_n^k &= \mathbf{u}_n^{k-1} + \delta \mathbf{u}_n^k, \\ \boldsymbol{\lambda}_n^k &= \boldsymbol{\lambda}_n^{k-1} + \delta \boldsymbol{\lambda}_n^k, \end{aligned} \quad \mathbf{q}_n^k = \left[\exp \left(\frac{1}{2} \mathcal{G}(\boldsymbol{\theta}_n^k) \right) \mathbf{p}_{n-1} \right]. \quad (22b)$$

Furthermore, the corrections in $\mathbf{v}_a$, i.e., $\delta \mathbf{v}_a$, can be substituted back into equations (22a) from the following relationships:

$$\delta \mathbf{v}_{a_n}^k = \hat{\mathbf{H}}(\boldsymbol{\theta}^{k-1}) \delta \mathbf{v}_n^k + \hat{\mathbf{W}}(\boldsymbol{\theta}^{k-1}, \dot{\boldsymbol{\theta}}^{k-1}) \delta \mathbf{u}_n^k, \quad \hat{\mathbf{W}} = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{W}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) \end{bmatrix}. \quad (23)$$

This step yields the following updated Newton-Raphson iteration:

$$\begin{aligned} \begin{bmatrix} \bar{\mathbf{M}}^{k-1} \delta \dot{\mathbf{v}}_n^k + \hat{\mathbf{F}}_{\text{Av}}^{k-1} \delta \mathbf{v}_n^k + \boldsymbol{\Gamma}_{\mathbf{u}}^{k-1} \delta \mathbf{u}_n^k + \mathbf{F}_{\text{A}\lambda}^{k-1} \delta \lambda_n^k \\ \boldsymbol{\Theta}(\boldsymbol{\Phi}^{k-1}) \delta \mathbf{u}_n^k \end{bmatrix} &= - \begin{bmatrix} \mathbf{d}_1^{k-1} \\ \mathbf{d}_2^{k-1} \end{bmatrix}, \\ \boldsymbol{\Gamma}_{\mathbf{u}}^{k-1} &= \left((\bar{\mathbf{M}} \dot{\mathbf{v}})^{k-1}_{\mathbf{u}} + \mathbf{F}_{\text{rot}\mathbf{u}}^{k-1} + \mathbf{F}_{\text{Av}_a}^{k-1} \hat{\mathbf{W}}(\boldsymbol{\theta}^{k-1}, \dot{\boldsymbol{\theta}}^{k-1}) + \boldsymbol{\Theta}(\mathbf{F}_{\text{A}}^{k-1}) \right), \\ \hat{\mathbf{F}}_{\text{Av}}^{k-1} &= \mathbf{F}_{\text{Av}_a}^{k-1} \hat{\mathbf{H}}(\boldsymbol{\theta}^{k-1}) + \mathbf{F}_{\text{rot}\dot{\mathbf{u}}}^{k-1}. \end{aligned} \quad (24)$$

In equations (24), it is imperative that the corresponding corrections in derivatives $\delta \mathbf{v}$ and $\delta \dot{\mathbf{v}}$ need to be expressed as some functions of $\delta \mathbf{u}$ and the step-size h . For instance, a simple first-order scheme [2, 3] simply translates to $\delta \mathbf{v} = \frac{1}{h} \delta \mathbf{u}$ and $\delta \dot{\mathbf{v}} = \frac{1}{h^2} \delta \mathbf{u}$. The problem at hand is to find the higher-order representations of the corrections $\delta \mathbf{v}$ and $\delta \dot{\mathbf{v}}$ to facilitate time-integration of order greater than 1. To achieve the same, the following subsection revisits an efficient TFE based family of integrators proposed by Kim and Reddy [29].

3.1 Direct implicit integration using TFE

This section proposes an extension of a TFE approach, originally made for structural dynamics problems [29], for solving the DAE of interest (10a). This approach illustrates a time finite element method based on modified weighted-residual method, applied to the velocity–displacement and velocity–acceleration relations. Consider the DAE (10a) re-written with separated algebraic and differential parts, as follows:

$$\begin{aligned} \begin{bmatrix} \bar{\mathbf{M}} \mathbf{a} + \hat{\mathbf{F}}_{\text{A}}(\mathbf{u}, \mathbf{v}, \lambda) \\ \frac{\boldsymbol{\Phi}(\mathbf{u})}{h^2} \end{bmatrix} &= \mathbf{0}, \quad \hat{\mathbf{F}}_{\text{A}} = \mathbf{F}_{\text{A}} + \mathbf{F}_{\text{rot}} \\ \dot{\mathbf{u}} &= \mathbf{v}, \\ \dot{\mathbf{v}} &= \mathbf{a}. \end{aligned} \quad (25)$$

The constraint equation is divided by h^2 for stabilizing the condition number of the resulting Jacobian matrix. It is important to notice that this form of DAE can be stated with $\mathbf{u}$ as a state variable because generalized coordinates $\mathbf{q}$ can be represented as a function of $\mathbf{u}$, as follows from equations (13) and (22b). Next, the variables $\mathbf{u}$ and λ are augmented into a single vector-valued variable $\mathbf{x} = [\mathbf{u}^T \quad \lambda^T]^T$. The corresponding augmented variables containing velocities and accelerations are denoted by $\mathbf{y} = \dot{\mathbf{x}} = [\mathbf{v}^T \quad \dot{\lambda}^T]^T$ and $\mathbf{z} = \ddot{\mathbf{x}} = [\dot{\mathbf{v}}^T \quad \ddot{\lambda}^T]^T$. The proposed TFE method is applied on the differential and algebraic parts by approximating independent signals $\mathbf{x}(t)$, $\mathbf{y}(t)$ and $\mathbf{z}(t)$ at time $t = t_{n-1} + \tau h$ within the interval $0 \leq \tau \leq 1$, using Lagrange polynomials $\varphi(\tau) = [\phi_0, \phi_1, \dots, \phi_m]$ of degree m , built on fixed set of interpolation points. These interpolation points, when considered in context of time discretization using finite elements, are essentially nodal coordinates within the time-element. For $m = 1, 2$, and 3 , these nodal coordinates, are tabulated in Table 1. It can be noticed that for

Table 1 Nodal coordinates in time-elements of different polynomial order

Polynomial degree (m)	Nodal coordinates
$m = 1$	$\{0, 1\}$
$m = 2$	$\{0, \frac{1}{2}, 1\}$
$m = 3$ (Equally-spaced)	$\{0, \frac{1}{3}, \frac{2}{3}, 1\}$
$m = 3$ (Gauss-Lobatto)	$\{0, \frac{1}{2} - \frac{\sqrt{5}}{10}, \frac{1}{2} + \frac{\sqrt{5}}{10}, 1\}$

$m = 3$, the distribution of nodal coordinates comes in two flavors: equidistant spacing and Gauss-Lobatto quadrature points.

The polynomial row vector $\varphi(\tau)$ contains $m + 1$ polynomial expressions associated with $m + 1$ nodes. The polynomial approximations of the augmented variables can be represented as follows:

$$\begin{aligned}\mathbf{x}(t) &= \bar{\varphi}(\tau)\bar{\mathbf{x}} + \phi_0(\tau)\mathbf{x}_{n-1}, & \bar{\varphi}(\tau) &:= [\phi_1, \dots, \phi_m] \otimes \mathbf{I}, \\ \mathbf{y}(t) &= \bar{\varphi}(\tau)\bar{\mathbf{y}} + \phi_0(\tau)\mathbf{y}_{n-1}, \\ \mathbf{z}(t) &= \bar{\varphi}(\tau)\bar{\mathbf{z}} + \phi_0(\tau)\mathbf{z}_{n-1}.\end{aligned}\quad (26)$$

Here, $\phi_0(\tau)$ is the polynomial associated with the first node that represents previous $(n - 1)$ time-step, where all the variables, i.e., $\mathbf{x}_{n-1}$, $\mathbf{y}_{n-1}$, $\mathbf{z}_{n-1}$, are known.

Hence, with m degree polynomials, each time finite element has $m + 1$ nodes with unknown variables associated with only m nodes. The vectors $\bar{\mathbf{x}}$, $\bar{\mathbf{y}}$, $\bar{\mathbf{z}} \in \mathbb{R}^{m(6N+C) \times 1}$, essentially comprise of unknowns $\mathbf{x}_i$, $\mathbf{y}_i$, $\mathbf{z}_i$ at i th node $\forall i \in 2, \dots, m + 1$. The differential parts with the augmented variables yield the following residual equations:

$$\mathbf{res}_1(t) = \dot{\mathbf{x}}(t) - \mathbf{y}(t), \quad \mathbf{res}_2(t) = \dot{\mathbf{y}}(t) - \mathbf{z}(t). \quad (27)$$

The modified weighted residual statements, as established in [29], applied on the isolated differential parts in equations (27) can be stated as follows:

$$\begin{aligned}\int_0^1 w(\tau) \frac{d^{j-1} \mathbf{res}_1(\tau)}{d\tau^{j-1}} d\tau &= \mathbf{0}, & \forall j &= 2, \dots, m + 1, \\ \int_0^1 w(\tau) \frac{d^{j-1} \mathbf{res}_2(\tau)}{d\tau^{j-1}} d\tau &= \mathbf{0}, & \forall j &= 2, \dots, m + 1.\end{aligned}\quad (28)$$

Evaluation of above integrals requires only a single weight and yet yields m linearly independent equations. Carrying on with the weight-optimization yields the following relations between the augmented variables:

$$\bar{\mathbf{y}} = \dot{\bar{\mathbf{x}}} = \bar{\alpha}\bar{\mathbf{x}} + \bar{\beta}\mathbf{x}_{n-1} + \bar{\gamma}\mathbf{y}_{n-1}, \quad (29a)$$

$$\bar{\mathbf{z}} = \dot{\bar{\mathbf{y}}} = \bar{\alpha}\bar{\mathbf{y}} + \bar{\beta}\mathbf{y}_{n-1} + \bar{\gamma}\mathbf{z}_{n-1}, \quad (29b)$$

$$\begin{aligned}\bar{\alpha} &= \bar{\alpha}_{m \times m} \otimes \mathbf{I}_{6N+C \times 6N+C}, \\ \bar{\beta} &= \bar{\beta}_{m \times 1} \otimes \mathbf{I}_{6N+C \times 6N+C}, \\ \bar{\gamma} &= \bar{\gamma}_{m \times 1} \otimes \mathbf{I}_{6N+C \times 6N+C}.\end{aligned}\quad (29c)$$

The matrices $\bar{\alpha}$, $\bar{\beta}$, $\bar{\gamma}$ are weight coefficient matrices, parameterized by $0 < \nu \leq 1$ as a damping controlling parameter, where $\nu = 1$ means no numerical damping. These matrices are

defined in Appendix B for $m \in \{1, 2, 3\}$. The subscripts in equations (29c) denote the size of the matrices and vectors, with m being the degree of polynomial used. The readers are advised to refer to [29] for more details on the derivation of weight matrices $\bar{\alpha}$, $\bar{\beta}$ and $\bar{\gamma}$. With ODEs, commonly countered in structural dynamics problems, the discussed TFE method ensures $2m - 1$ order of the solution [29]. Therefore, it is of primary interest to study this family of TFE integrators and the effect of nodal distributions in time-elements in the contexts of DAEs.

3.2 Solution strategy using Newton-Raphson method

Consider the time finite element with $m + 1$ nodes. Node $i = 1$ corresponds to the time-step $t_0 = t_{n-1}$. Hence, the state variables $\mathbf{u}_{n-1}$, $\mathbf{v}_{n-1}$, $\dot{\mathbf{v}}_{n-1}$, $\boldsymbol{\lambda}_{n-1}$ are known values at node $i = 1$. To find the unknowns at the i th node, $\forall i = 2, 3, \dots, m + 1$, the Newton-iteration (22a) can be stated in terms of augmented variables $\mathbf{x}_i = [\mathbf{u}_i^T, \boldsymbol{\lambda}_i^T]^T$ as follows:

$$\begin{aligned} -\mathbf{d}_i^{k-1} &= \begin{bmatrix} \bar{\mathbf{M}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}_i^{k-1} \begin{bmatrix} \delta \dot{\mathbf{v}} \\ \delta \ddot{\boldsymbol{\lambda}} \end{bmatrix}_i^k + \begin{bmatrix} \hat{\mathbf{F}}_{\text{Av}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}_i^{k-1} \begin{bmatrix} \delta \mathbf{v} \\ \delta \dot{\boldsymbol{\lambda}} \end{bmatrix}_i^k + \begin{bmatrix} \boldsymbol{\Gamma}_{\mathbf{u}} & \mathbf{F}_{\Lambda\lambda} \\ \boldsymbol{\Theta}(\Phi)/h^2 & \mathbf{0} \end{bmatrix}_i^{k-1} \begin{bmatrix} \delta \mathbf{u} \\ \delta \boldsymbol{\lambda} \end{bmatrix}_i^k, \\ \Rightarrow -\mathbf{d}_i^{k-1} &= \tilde{\mathbf{M}}_i \delta \ddot{\mathbf{x}}_i^k + \tilde{\mathbf{C}}_i^{k-1} \delta \dot{\mathbf{x}}_i^k + \tilde{\mathbf{K}}_i^{k-1} \delta \mathbf{x}_i^k. \end{aligned} \quad (30)$$

The equations (30) for all the nodes in the system can be assembled together to yield the following Newton-Raphson equations:

$$\begin{bmatrix} \tilde{\mathbf{M}}_2 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{M}}_3 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \tilde{\mathbf{M}}_{m+1} \end{bmatrix}^{k-1} \begin{bmatrix} \delta \ddot{\mathbf{x}}_2 \\ \delta \ddot{\mathbf{x}}_3 \\ \vdots \\ \delta \ddot{\mathbf{x}}_{m+1} \end{bmatrix}^k + \begin{bmatrix} \tilde{\mathbf{C}}_2 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{C}}_3 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \tilde{\mathbf{C}}_{m+1} \end{bmatrix}^{k-1} \begin{bmatrix} \delta \dot{\mathbf{x}}_2 \\ \delta \dot{\mathbf{x}}_3 \\ \vdots \\ \delta \dot{\mathbf{x}}_{m+1} \end{bmatrix}^k \quad (31a)$$

$$\begin{aligned} &+ \begin{bmatrix} \tilde{\mathbf{K}}_2 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{K}}_3 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \tilde{\mathbf{K}}_{m+1} \end{bmatrix}^{k-1} \begin{bmatrix} \delta \mathbf{x}_2 \\ \delta \mathbf{x}_3 \\ \vdots \\ \delta \mathbf{x}_{m+1} \end{bmatrix}^k = - \begin{bmatrix} \mathbf{d}_2 \\ \mathbf{d}_3 \\ \vdots \\ \mathbf{d}_{m+1} \end{bmatrix}^{k-1}, \\ \Rightarrow \bar{\mathbf{M}} \delta \ddot{\mathbf{x}}^k + \bar{\mathbf{C}}^{k-1} \delta \dot{\mathbf{x}}^k + \bar{\mathbf{K}}^{k-1} \delta \mathbf{x}^k &= -\bar{\mathbf{d}}^{k-1}. \end{aligned} \quad (31b)$$

Using the equations (29a)–(29c), the corrections in accelerations and velocities at all nodes can be represented as follows:

$$\delta \dot{\mathbf{x}} = \bar{\boldsymbol{\alpha}} \delta \ddot{\mathbf{x}}, \quad \delta \ddot{\mathbf{x}} = \bar{\boldsymbol{\alpha}} \delta \dot{\mathbf{x}} = \bar{\boldsymbol{\alpha}}^2 \delta \mathbf{x}. \quad (32)$$

Plugging the relations from equation (32) into (31b), the final Newton-Raphson iteration with TFE method can be retrieved as follows:

$$[\bar{\mathbf{M}} \bar{\boldsymbol{\alpha}}^2 + \bar{\mathbf{C}}^{k-1} \bar{\boldsymbol{\alpha}} + \bar{\mathbf{K}}]^{k-1} \delta \mathbf{x}^k = -\bar{\mathbf{d}}^{k-1}, \quad (33a)$$

Algorithm 1 TFE integration of multibody dynamics with friction in absolute coordinates

```

1: Inputs:  $m, \nu, t_0, t_{\text{final}}, \lambda_0, \mathbf{q}_0, \mathbf{v}_0, \dot{\mathbf{v}}_0$ 
2:  $n \leftarrow 0$ 
3:  $\boldsymbol{\theta}_0 \leftarrow \mathbf{0}$ 
4: Call  $\bar{\boldsymbol{\alpha}}, \bar{\boldsymbol{\beta}}, \bar{\boldsymbol{\gamma}}$  for selected  $m$  and  $\nu$ 
5: while  $t_n < t_{\text{final}}$  do
6:    $k \leftarrow 1$ 
7:   Evaluate guess states  $\ddot{\mathbf{x}}^k, \dot{\mathbf{x}}^k, \bar{\mathbf{x}}^k$  ▷ Using (34)
8:   while  $\|\bar{\mathbf{d}}\| \geq \varepsilon$  do
9:     for  $j = 2 : m + 1$  do
10:      Extract nodal states  $\mathbf{v}_j, \mathbf{u}_j, \lambda_j, \dot{\mathbf{v}}_j$  from  $\ddot{\mathbf{x}}^k, \dot{\mathbf{x}}^k, \bar{\mathbf{x}}^k$ 
11:      Evaluate  $\mathbf{q}_j$  ▷ Using (15)
12:      Evaluate nodal entities  $\tilde{\mathbf{M}}_j, \tilde{\mathbf{C}}_j, \tilde{\mathbf{K}}_j$  ▷ Using (30)
13:    end for
14:    Assemble  $\bar{\mathbf{M}}, \bar{\mathbf{C}}, \bar{\mathbf{K}}, \bar{\mathbf{d}}$  ▷ Using (31a)–(31b)
15:    Solve  $k$ th Newton-Raphson iteration ▷ Using (33a)
16:    Update the global state-ensemble ▷ Using (33b)
17:     $k \leftarrow k + 1$ 
18:  end while
19:  Extract the states at  $(m + 1)^{\text{th}}$  node
20:   $\begin{bmatrix} \ddot{\mathbf{x}}_{n-1} \\ \dot{\mathbf{x}}_{n-1} \\ \bar{\mathbf{x}}_{n-1} \end{bmatrix} \leftarrow \begin{bmatrix} \ddot{\mathbf{x}}_n \\ \dot{\mathbf{x}}_n \\ \bar{\mathbf{x}}_n \end{bmatrix}$ 
21:   $t_n \leftarrow t_{n-1} + h$ 
22:   $n \leftarrow n + 1$ 
23: end while

```

$$\begin{aligned}
\bar{\mathbf{x}}^k &= \bar{\mathbf{x}}^{k-1} + \delta \bar{\mathbf{x}}^k, \\
\dot{\mathbf{x}}^k &= \bar{\boldsymbol{\alpha}} \bar{\mathbf{x}}^k + \bar{\boldsymbol{\beta}} \mathbf{x}_{n-1} + \bar{\boldsymbol{\gamma}} \dot{\mathbf{x}}_{n-1}, \\
\ddot{\mathbf{x}}^k &= \bar{\boldsymbol{\alpha}} \dot{\mathbf{x}}^k + \bar{\boldsymbol{\beta}} \dot{\mathbf{x}}_{n-1} + \bar{\boldsymbol{\gamma}} \ddot{\mathbf{x}}_{n-1}, \\
\mathbf{q}_i^k &= \begin{bmatrix} \mathbf{r}_i^k \\ \exp\left(\frac{1}{2} \mathcal{G}(\boldsymbol{\theta}_i^k)\right) \mathbf{p}_{n-1} \end{bmatrix}, \quad \forall i = 2, 3, \dots, m + 1.
\end{aligned} \tag{33b}$$

The initial guess values at $k = 1$ are chosen using the following equations:

$$\begin{aligned}
\ddot{\mathbf{x}}^{k=1} &= \ddot{\mathbf{x}}_{n-1}, \\
\dot{\mathbf{x}}^{k=1} &= \dot{\mathbf{x}}_{n-1} + \tau \ddot{\mathbf{x}}_{n-1}, \\
\bar{\mathbf{x}}^{k=1} &= \bar{\mathbf{x}}_{n-1} + \tau \dot{\mathbf{x}}_{n-1}.
\end{aligned} \tag{34}$$

A step-by-step procedure to implement the discussed index-3 DAE formulation with TFE integration on Lie-group matrices of quaternions, is described in the Algorithm 1. For a target simulation time t_{final} , the initial values of $\boldsymbol{\theta}_0$ at time t_0 should be assumed zero.

The extracted states on $(m + 1)^{\text{th}}$ node, i.e., at $\tau = 1$, represent the states at the n th time step in step 18 of the algorithms 1. These can be later used to evaluate contact forces and friction forces and the torques generated by them using the equations discussed in [5]. It is

worth noting that presented TFE schemes not only promise higher order accuracy, i.e., upto 5th order as discussed in this paper, but also feature a strong potential for parallelization. Essentially, steps 9 to 12 in algorithm 1 can be executed in a parallel computing environment that can improve the computational performance, i.e., the nodal entities corresponding to each node can be computed parallel to each other and then combined in step 14. However, the topic of parallelization is not aligned with the present scope of this study, although it is worth investigating as future study.

3.3 Implementation of Lie-group Newmark- β and trapezoidal schemes

The generalized definition of operator Θ proposed in Sect. 2.2 also allows implementation of conventional integration schemes. This subsection discusses implementation of two commonly used schemes in multibody applications, namely, Trapezoidal rule and Newmark- β method. Their implementation with proposed formulation also lays groundwork for assessing the performance of TFE integration in context of DAEs. No such analyses have been reported in existing literature, to the best of our knowledge.

Consider the DAE (25) with separated algebraic and differential state variables. The trapezoidal rule further yields following equations:

$$\mathbf{v}_n = \mathbf{v}_{n-1} + \frac{h}{2}(\mathbf{a}_{n-1} + \mathbf{a}_n), \quad (35a)$$

$$\mathbf{u}_n = \mathbf{u}_{n-1} + \frac{h}{2}(\mathbf{v}_{n-1} + \mathbf{v}_n). \quad (35b)$$

Since the DAE (25) is inherently a set of nonlinear algebraic equations, a Newton-Raphson scheme is inevitable to solving the nonlinear system. The resulting linear system for evaluating corrections $\delta \mathbf{y}^k$ at the k th Newton-Raphson iteration, can thus be stated as follows:

$$\bar{\mathbf{J}}^{k-1} \delta \mathbf{y}^k = -\mathbf{res}^{k-1}, \quad (36a)$$

$$\mathbf{res}^{k-1} = \begin{bmatrix} \bar{\mathbf{M}}\mathbf{a}^{k-1} + \hat{\mathbf{F}}_{\mathbf{A}}(\mathbf{u}^{k-1}, \mathbf{v}^{k-1}, \boldsymbol{\lambda}^{k-1}) \\ -\mathbf{v}^{k-1} + \mathbf{v}_{n-1} + \frac{h}{2}(\mathbf{a}_{n-1} + \mathbf{a}^{k-1}) \\ -\mathbf{u}^{k-1} + \mathbf{u}_{n-1} + \frac{h}{2}(\mathbf{v}_{n-1} + \mathbf{v}^{k-1}) \\ \Phi(\mathbf{u}^{k-1})/h^2 \end{bmatrix}, \quad \delta \mathbf{y}^k = \begin{bmatrix} \delta \mathbf{a}^k \\ \delta \mathbf{v}^k \\ \delta \mathbf{u}^k \\ \delta \boldsymbol{\lambda}^k \end{bmatrix}, \quad (36b)$$

$$\bar{\mathbf{J}}^{k-1} = \begin{bmatrix} \bar{\mathbf{M}} & \hat{\mathbf{F}}_{\mathbf{A}\mathbf{v}} & \boldsymbol{\Gamma}_{\mathbf{u}} & \mathbf{F}_{\mathbf{A}\boldsymbol{\lambda}} \\ \frac{h}{2}\mathbf{I} & -\mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \frac{h}{2}\mathbf{I} & -\mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Theta(\Phi)/h^2 & \mathbf{0} \end{bmatrix}^{k-1}.$$

Although the trapezoidal rule has been well established for problems with friction [34] using tangent-space index-0 DAE, the periodic components contributed by Θ operator can influence the stability of solution with the formulation proposed in this paper. The trapezoidal integration rule comes with equal contribution from explicit and implicit parts. However, the lack of controllable numerical damping in trapezoidal rule often makes it unsuitable for problems with high-frequency periodic noise.

The Newmark- β method is a popular integrator of choice for structural and multibody dynamics equations, and can be derived from generalized- α method by keeping $\alpha = 0$ [35–37].

Algorithm 2 Newmark- β and trapezoidal integration for multibody dynamics with friction in absolute coordinates

```

1: Inputs:  $t_0, t_{\text{final}}, \beta, \gamma, \lambda_0, \mathbf{q}_0, \mathbf{v}_0, \mathbf{a}_0$ 
2:  $n \leftarrow 0$ 
3:  $\boldsymbol{\theta}_0 \leftarrow \mathbf{0}$ 
4: while  $t_n < t_{\text{final}}$  do
5:    $k \leftarrow 1$ 
6:   Initialize  $\mathbf{y}^k \leftarrow \mathbf{y}_n$ 
7:   Evaluate  $\mathbf{q}^k$  ▷ Using (15)
8:   while  $\|\mathbf{res}\| \geq \varepsilon$  do
9:     Evaluate  $\bar{\mathbf{J}}(\mathbf{y}^k, \mathbf{q}^k), \mathbf{res}(\mathbf{y}^k, \mathbf{q}^k)$  ▷ Using (36a)–(36b) or (38)
10:     $\mathbf{y}^k \leftarrow \mathbf{y}^k - \bar{\mathbf{J}}^{-1} \mathbf{res}$ 
11:    Evaluate  $\mathbf{q}^k$  ▷ Using (15)
12:     $k \leftarrow k + 1$ 
13:  end while
14:   $\mathbf{y}_n \leftarrow \mathbf{y}^k$ 
15:   $t_n \leftarrow t_{n-1} + h$ 
16:   $n \leftarrow n + 1$ 
17: end while

```

Unlike, the trapezoidal rule, this method is based on two additional parameters: γ to control order and β to control numerical damping. This scheme ensures second-order accuracy with $\gamma = \frac{1}{2}$ and is equivalent to trapezoidal rule with $\gamma = \frac{1}{2}$ and $\beta = \frac{1}{4}$, i.e. no numerical damping. However, numerical damping can be induced by changing β [35]. The Newmark scheme yields the following equations:

$$\mathbf{v}_n = \mathbf{v}_{n-1} + h((1 - \gamma)\mathbf{a}_{n-1} + \gamma\mathbf{a}_n), \quad (37a)$$

$$\mathbf{u}_n = \mathbf{u}_{n-1} + h\mathbf{v}_{n-1} + h^2\left(\left(\frac{1}{2} - \beta\right)\mathbf{a}_{n-1} + \beta\mathbf{a}_n\right). \quad (37b)$$

For $\beta \neq 0$, the above equations are implicit and so is the DAE (25). The resulting Newton-Raphson iteration, has the following elements:

$$\mathbf{res}^{k-1} = \begin{bmatrix} \bar{\mathbf{M}}\mathbf{a}^{k-1} + \hat{\mathbf{F}}_{\Lambda}(\mathbf{u}^{k-1}, \mathbf{v}^{k-1}, \boldsymbol{\lambda}^{k-1}) \\ -\mathbf{v}^{k-1} + \mathbf{v}_{n-1} + h((1 - \gamma)\mathbf{a}_{n-1} + \gamma\mathbf{a}^{k-1}) \\ -\mathbf{u}^{k-1} + \mathbf{u}_{n-1} + h\mathbf{v}_{n-1} + h^2\left(\left(\frac{1}{2} - \beta\right)\mathbf{a}_{n-1} + \beta\mathbf{a}^{k-1}\right) \\ \boldsymbol{\Phi}(\mathbf{u}^{k-1})/h^2 \end{bmatrix}, \quad (38)$$

$$\bar{\mathbf{J}}^{k-1} = \begin{bmatrix} \bar{\mathbf{M}} & \hat{\mathbf{F}}_{\Lambda\mathbf{v}} & \boldsymbol{\Gamma}_{\mathbf{u}} & \mathbf{F}_{\Lambda\lambda} \\ \gamma h \mathbf{I} & -\mathbf{I} & \mathbf{0} & \mathbf{0} \\ h^2 \beta \mathbf{I} & \mathbf{0} & -\mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Theta}(\boldsymbol{\Phi})/h^2 & \mathbf{0} \end{bmatrix}^{k-1}.$$

Algorithm 2 delineates a step-by-step approach for implementing trapezoidal and Newmark- β schemes with the proposed formulation.

Fig. 1 Schematic diagram of a pendulum demonstrating a revolute pair

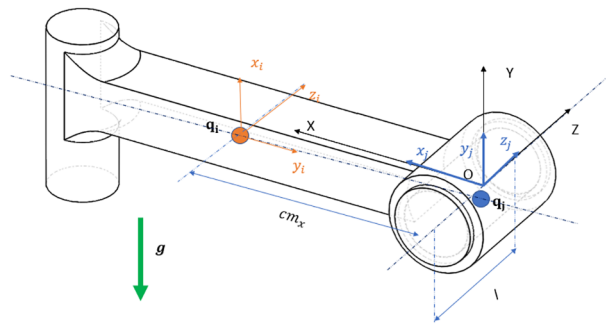


Table 2 Case study specifications: Revolute pair

Property	Value	Units
m	10	kg
$\mathbf{J}$	$\begin{bmatrix} 0.05 & 0 & 0 \\ 0 & 0.03 & 0.015 \\ 0 & 0.015 & 0.028 \end{bmatrix}$	kg-m ²
l	2	m
R	0.5	m
cm_x	3.09	m
μ_s	0.3	—
μ_d	0.2	—

4 Numerical experiments

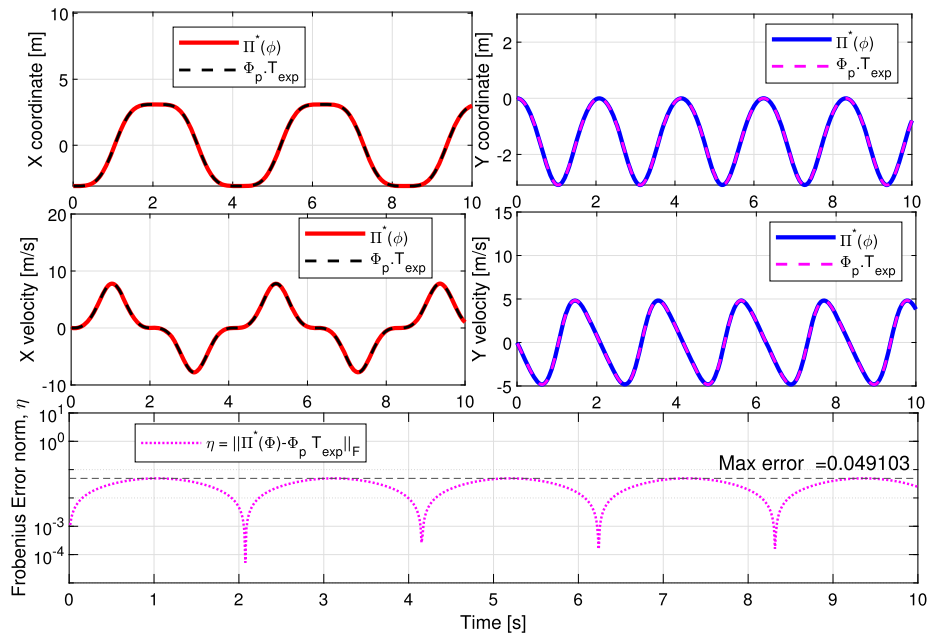
The test cases are based on the system in Fig. 1, where a pendulum i forms a revolute joint with a hinge pin j of radius R . The center of mass of the pendulum is located at cm_x along local X direction. The pendulum i has mass m and inertia matrix $\mathbf{J}$ with its center of mass denoted by the generalized coordinates $\mathbf{q}_i \in \mathbb{R}^7$ and the hinge pin j is mass-less and fixed at both ends. The gravity acts along the $-Y$ axis. The system specific details illustrated in Fig. 1 are shown in Table 2.

Two cases of the pendulum are simulated with all the discussed integrators: (1) frictionless pendulum, and (2) pendulum with friction in the revolute joint with static and dynamic coefficients μ_s and μ_d respectively. The friction model used in this study was Brown and McPhee's velocity based continuous friction model [38]. For more details on continuous friction models, readers are advised to refer to preliminary work by authors [39]. Furthermore, all the partial derivatives/ Jacobian matrices were computed using a central finite difference scheme with perturbation value $\sqrt{\bar{\epsilon}}$, where $\bar{\epsilon}$ is machine precision value. The damping control parameters for the compared integrators are tabulated in Table 3.

As a proof of concept, Fig. 2 shows the Frobenius error norm defined in equation (19). Both methods produce same solution and the Frobenius error is observed to have an upper bound 0.049. Figures 3 and 4 compare TFE ($m = 1, 2, 3$), Newmark- β and trapezoidal integrators for two step-sizes, $h = 0.003$ and $h = 0.006$ respectively, in case of a frictionless pendulum. A quick observation is the instability of the trapezoidal method with both step-sizes. This can be observed from the constraint force plots as well as the total mechanical energy plot. Among TFE family, the energy conservation improves marginally with increasing the degree of polynomial. The Newmark- β scheme performs better than TFE ($m = 1$)

Table 3 Damping control parameters of integrators under comparison

Integrator	Damping parameter	Friction neglected	Friction considered
TFE ($m = 1$)	ν	0.99	0.99
TFE ($m = 2$)	ν	0.95	0.95
TFE ($m = 3$)	ν	0.9	0.9
Trapezoidal	—	—	—
Newmark- β ($\gamma = 0.5$)	β	0.3	0.3

**Fig. 2** Frictionless pendulum: Equivalence of proposed definition of the operator Π in equations (17) to that proposed in [1–3]. Frobenius norm computed as per equation (19)

in terms of mitigating energy deviation. However, TFE for $m = 2, 3$ further reduce the energy deviation. In all cases, the constraints were found to be well-preserved within machine precision. Furthermore, the iteration plots show that with step-size $h = 0.006$, all the TFE methods took 3 iterations at each time-step to achieve convergence criteria of $\varepsilon = 1e - 7$, while the iteration count for Newmark scheme varied periodically between 3 and 4. With step-size $h = 0.003$, TFE($m = 3$) intermittently required 4 iterations, while TFE($m = 1$ and $m = 2$) converged within 3 iterations per time-step.

Figure 6 compares the evolution of error and the obtained solution order in coordinates $\mathbf{q}$ and velocities $\mathbf{v}$ with increasing step-sizes. Although [29] guarantee $2m - 1$ order, reduction in order can be noticed in TFE implementation for DAEs for $m = 2, 3$. In fact, it is observed that TFE ($m = 3$) achieves only fourth order accuracy. The order-loss could be results of multiple contributors. The prominent ones include inherent stiffness of DAE formulation, as opposed to ODE formulation, and the truncation index in (6a) being limited to the third

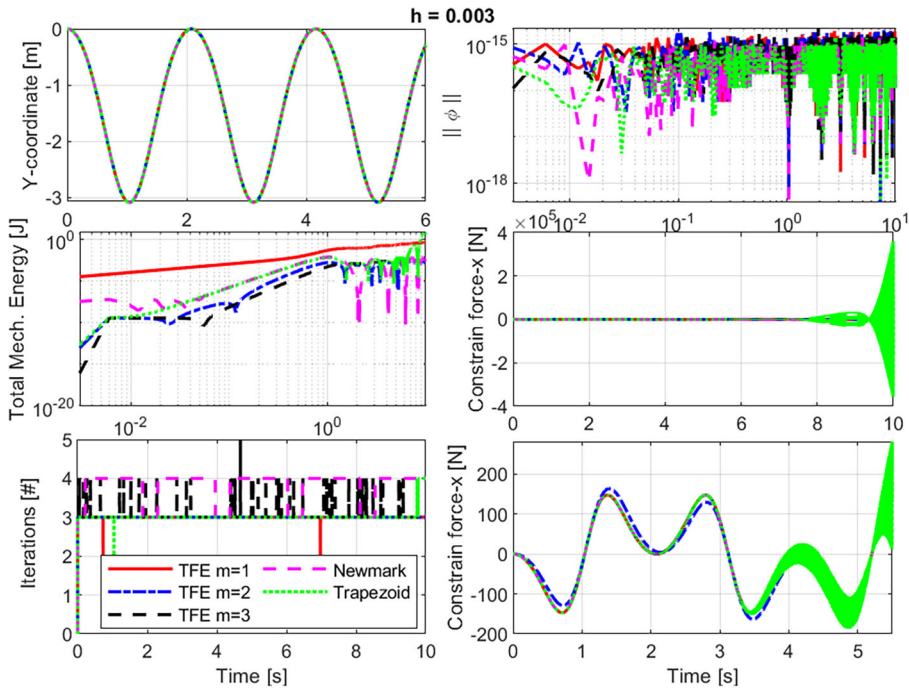


Fig. 3 Frictionless pendulum: Comparison of Newmark and Trapezoidal schemes with TFE with step-size $3e-3$. Trapezoidal scheme gives unstable solution

term [12]. A formal investigation into this problem is hence an immediate future scope of this problem.

To evaluate errors for TFE and Newmark methods with step-sizes in range $6e-3 \leq h \leq 1.2e-2$, the reference solution was obtained for the respective integrator with step-size $1e-4$. Figure 5 shows computational cost of all the integrators compared. Because of increase in dimensionality, it is expected for TFE schemes with $m=2$ and $m=3$ to consume for computational resources. However, juxtaposing the Figs. 6 and 5, it can be noticed that using higher degree of polynomial allows using bigger step-size to achieve same order of error as with lower degree of polynomial. For instance, using TFE with $m=3$, $h=1.2e-2$ yields error $< 1e-3$ at $t_{final}=10s$ for the simulation time $75s$, while $m=2$, $h=6e-3$ yields error of order $1e-3$ at $t_{final}=10s$ for the simulation time of $81s$. Furthermore, Newmark- β scheme consistently gives second-order accurate solutions with lowest computational effort. This can be partially attributed to the lowest dimensionality of the resulting Newton-iteration, only next to TFE ($m=1$).

Figures 7, 8, and 9 illustrate the evolving deviation in mechanical energy, error propagation in η , from equation (19), and the number of Newton iterations to convergence per time-step (bottom-most plot), for TFE with $m=1, 2, 3$ respectively. It can be noticed that Newmark- β takes the highest number of total iterations. Figure 10 shows the same quantities for Newmark- β scheme. The work-precision diagrams, shown in Fig. 11 summarize the key-inferences from Figs. 6–11. It is evident that with $m=3$, higher accuracy comes with increased computational cost. It can be observed that increasing the order of polynomial indeed mitigates the error propagation, however the total number of Newton-iterations by the end of t_{final} do not differ significantly.

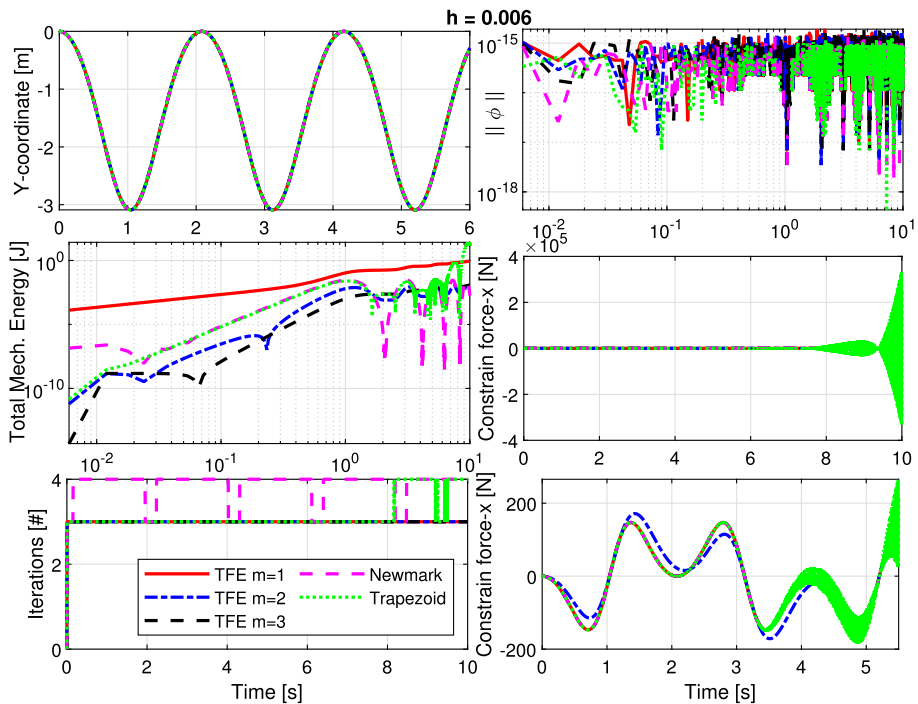


Fig. 4 Frictionless pendulum: Comparison of Newmark and Trapezoidal schemes with TFE with step-size $6e-3$. Trapezoidal scheme gives unstable solution

Figures 12 and 13 present results for simulations with friction accounted in the revolute joint with $\mu_s = 0.3$ and $\mu_d = 0.2$, for step-sizes $h = 0.003$ and $h = 0.008$ respectively. Just like in the frictionless case, the trapezoidal rule fails to produce a stable solution. Furthermore, a quick observation shows that TFE ($m = 3$) requires more iterations at lower step-size $h = 3e-3$ and exhibits a stable range of Newton iterations on large step-sizes. This correlates with the relatively high condition number of Newton-Raphson matrix for $m = 3$ as shown in Fig. 14. Unlike the case of frictionless pendulum, the TFE schemes take more iterations (5–6) than Newmark scheme for problems with friction.

Concluding the analysis, solutions obtained from equally-spaced nodes and Gauss-Lobatto nodal points are summarized in Fig. 15. For same damping parameter, TFE ($m = 3$) fails to produce a stable solution with equally-spaced nodes for all values of ν in this particular experiment. Figure 16 shows the total simulation time with friction considered. It can be observed that adding nonlinearity such as friction may increase the computational resources required to achieve convergence for TFE schemes, while Newmark- β and trapezoidal schemes show marginal increment in simulation time. Furthermore, Fig. 17 compares the evolution of error and the obtained solution order in coordinates $\mathbf{q}$ and velocities $\mathbf{v}$ with increasing step-sizes. Compared to friction-less case, accounting-in friction resulted in greater order loss, and larger error values. In addition, there is marginal improvement in errors in position with the usage of higher order methods, when compared to the first-order solution. However, a clear advantage of TFE schemes can be observed from Fig. 18, where TFE ($m=2$) is compared with Newmark- β at time-step $h = 0.2$. The solution from TFE scheme, with friction taken into account, yields mechanical energy at $t_{final} = 10s$ close to

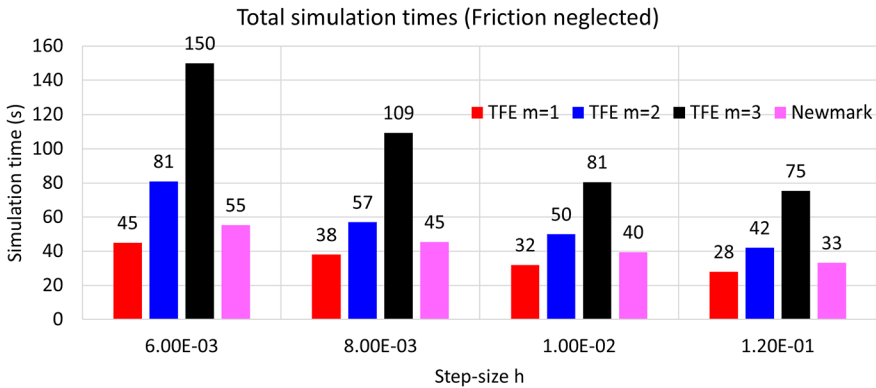


Fig. 5 Frictionless pendulum: Computational times for a 10 s simulation

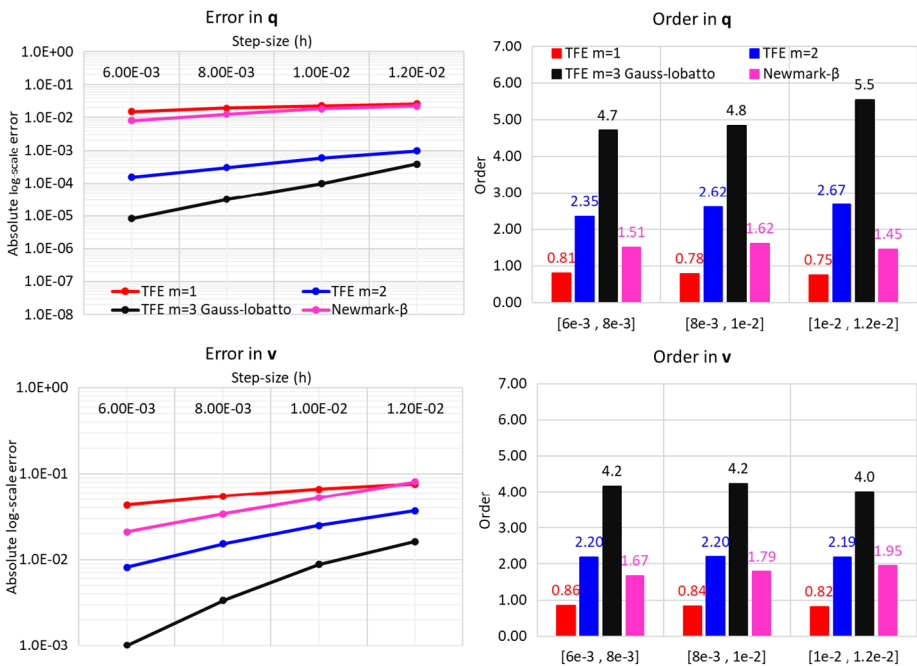


Fig. 6 Frictionless pendulum: Solution order of q and v with TFE and Newmark- β schemes. Reference solution obtained with step-size $h = 1e - 4$, for each integrator

the correct solution. On the other hand, Newmark- β fails to achieve convergence and yields incorrect solution.

5 Conclusions

In popular approaches for simulating multibody systems, rotation matrices of all the bodies are expressed as functions of their global rotational coordinates such as unit-quaternions

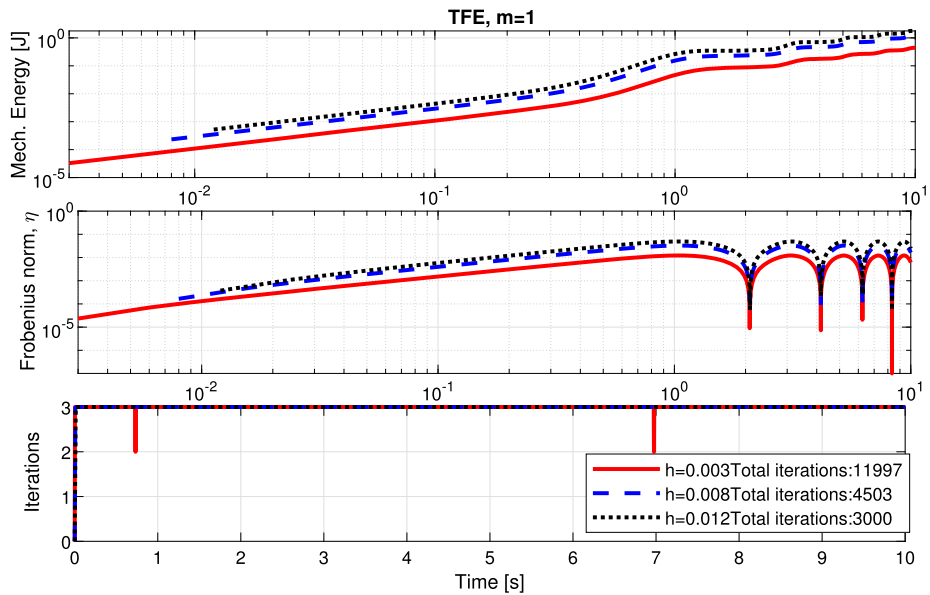


Fig. 7 Frictionless pendulum: Propagation of errors with polynomial order $m = 1$. Top panel: mechanical energy, Middle panel: Frobenius error norm η , from equation (19), Bottom panel: Newton iterations

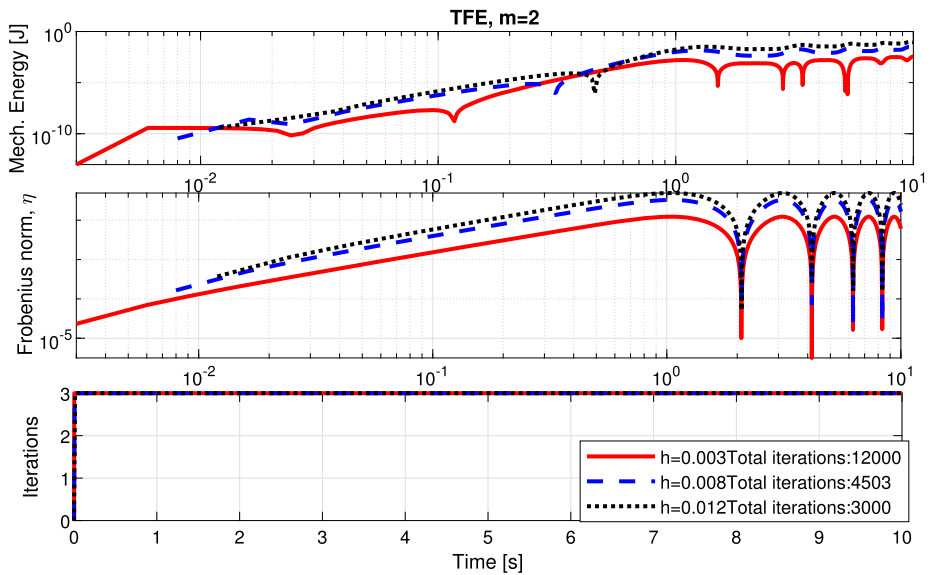


Fig. 8 Frictionless pendulum: Propagation of errors with polynomial order $m = 2$. Top panel: mechanical energy, Middle panel: Frobenius error norm η , from equation (19), Bottom panel: Newton iterations

(Euler parameters) or Euler angles at each time step. Though unit-quaternions can circumvent the problem of gimbal lock, associated with Euler parameters, they carry an additional normalization constraint. The methodology proposed in this contribution uses direct expo-

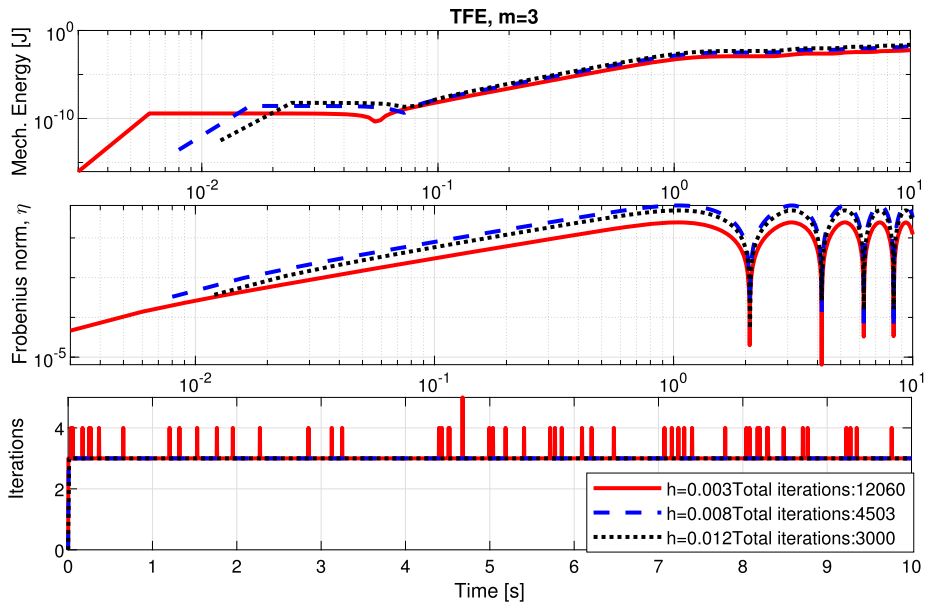


Fig. 9 Frictionless pendulum: Propagation of errors with polynomial order $m = 3$. Top panel: mechanical energy, Middle panel: Frobenius error norm η , from equation (19), Bottom panel: Newton iterations

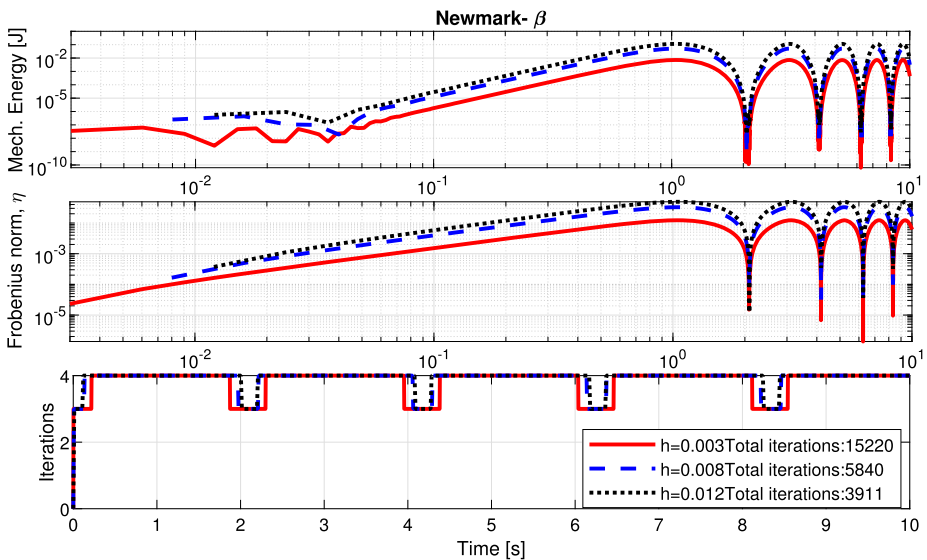


Fig. 10 Frictionless pendulum: Propagation of errors with Newmark- β scheme. Top panel: mechanical energy, Middle panel: Frobenius error norm η , from equation (19), Bottom panel: Newton iterations

nential integration on S^3 to eschew the additional constraint. The methodology discussed herein contributes an approach of systematically formulating the first-order variations of quaternions in terms of variation in body-frame Euler-rotations. This is useful in convert-

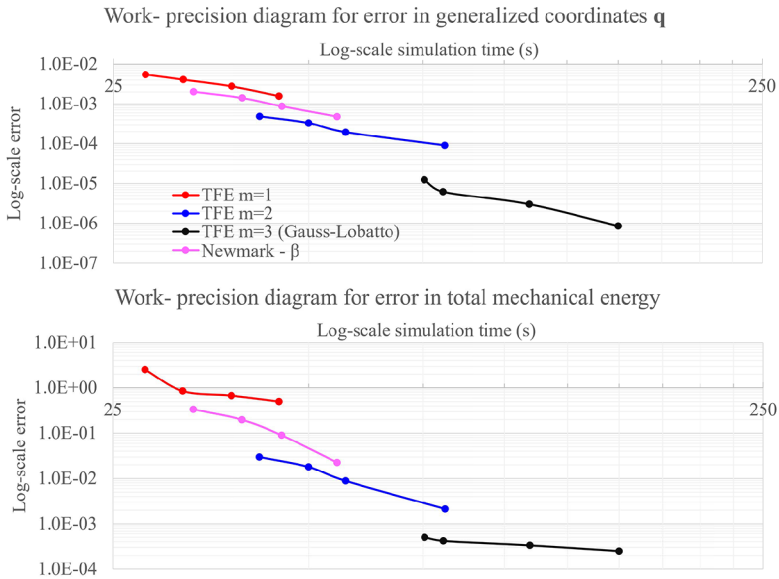


Fig. 11 Frictionless pendulum: Work-precision diagrams for TFE and Newmark- β schemes

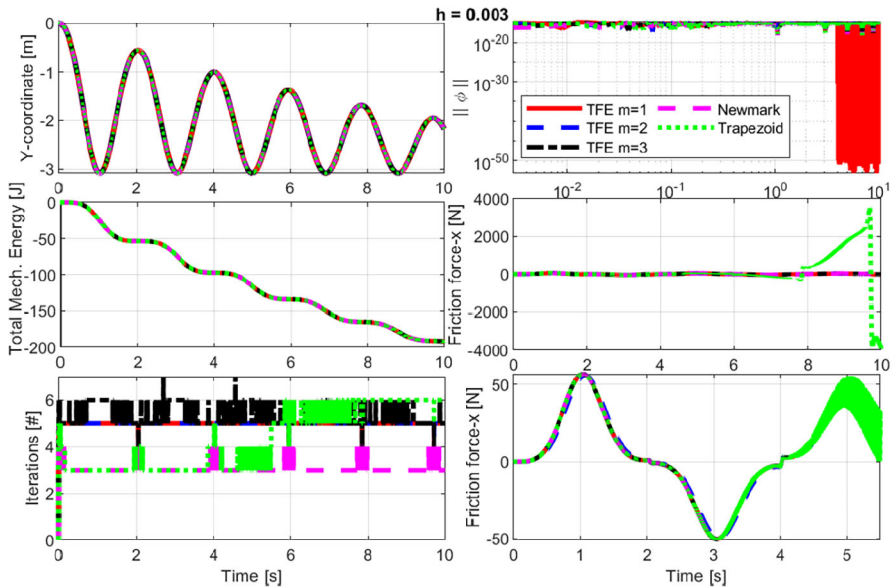


Fig. 12 Pendulum with friction ($\mu_s = 0.3$, $\mu_d = 0.2$): Comparison of Newmark and Trapezoidal schemes with TFE with step-size $3e - 3$. Trapezoidal scheme gives unstable solution

ing the first-order variation in an arbitrary nonlinear force function with respect to Euler parameters, to that with respect to body-frame rotations. Since the angular velocity is a di-

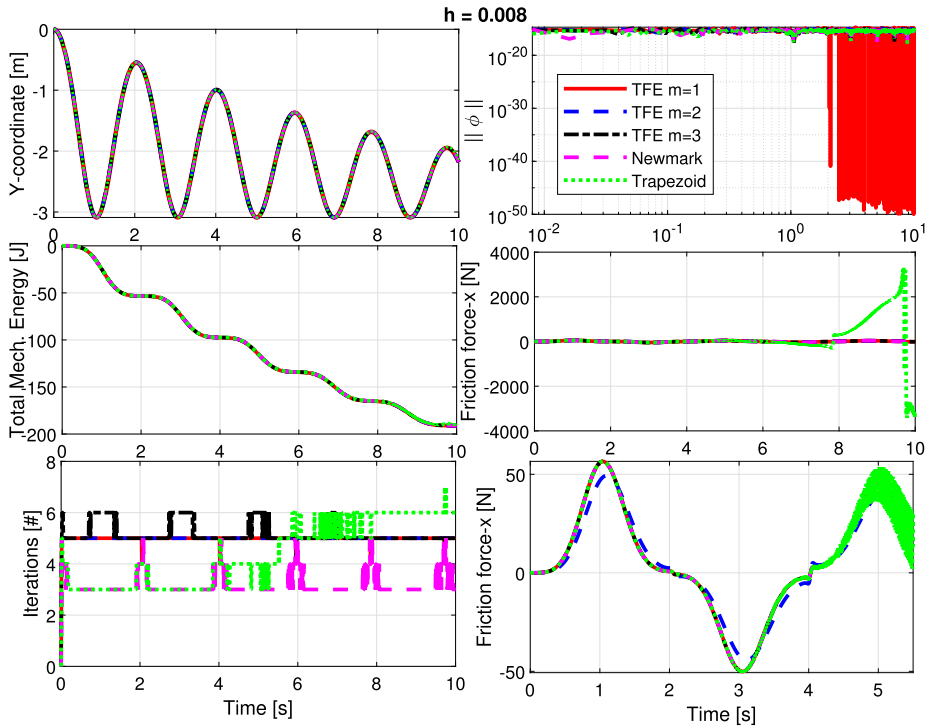


Fig. 13 Pendulum with friction ($\mu_s = 0.3$, $\mu_d = 0.2$): Comparison of Newmark and Trapezoidal schemes with TFE with step-size $8e-3$. Trapezoidal scheme gives unstable solution

rect derivative of these Euler-rotations, this approach enables higher-order integration using Newton-Raphson iteration.

To facilitate higher-order integration, time finite element methods are explored on quaternion Lie groups. Due to advantages of easy computer implementation, a TFE approach facilitating direct integration [29] was used. The TFE methods were modified to be used with the index-3 DAE by augmenting the Lagrange multipliers with position and orientation variables. Three integrators corresponding to degree of approximating polynomials, m , were used in this study, i.e., $m = 1, 2$ and 3 . In addition to this, implementation of a Newmark- β scheme and a trapezoidal scheme on quaternion Lie-groups was presented and the TFE schemes were compared with the later two. The trapezoidal rule yielded unstable solution across the range of step-sizes, while Newmark- β scheme produced stable results with a positive damping, i.e., $\beta = 0.3$.

For analyzing error characteristics of TFE and Newmark schemes, the reference solution was obtained with step-size $1e-4$ for each integration scheme. While TFE schemes were found to yield better error control on energy-deviation even with large-step sizes, the approach has limitations in terms of computational cost when nonlinearity, such as friction, is added to the system. However, for frictionless case, it was observed that this cost can be compensated by using higher degree polynomial with larger step-sizes. Furthermore, since the number of Newton-iterations for frictionless pendulum were the same for all integrators, the additional computational time for $m = 2, 3$ can be attributed to the dimensionality of the resulting Newton-iteration equations. Yet, in the case with friction accounted in, the TFE schemes exhibited much larger increase in simulation time than trapezoidal and Newmark- β

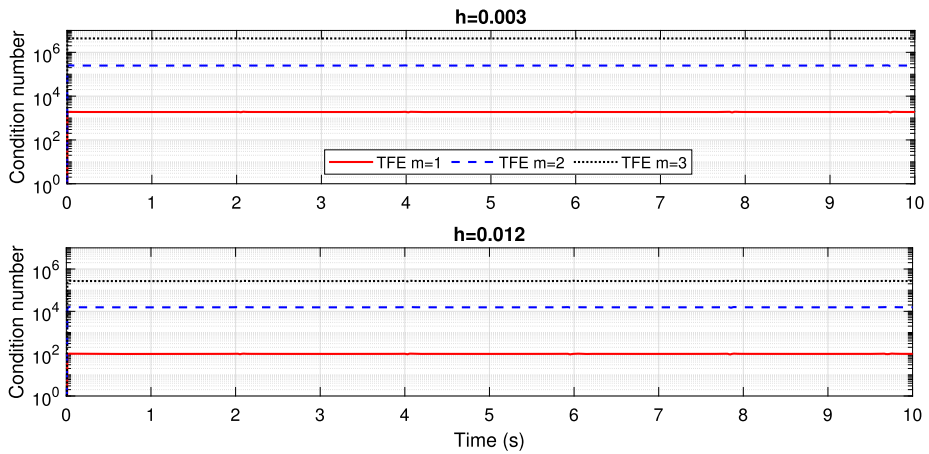


Fig. 14 Pendulum with friction ($\mu_s = 0.3$, $\mu_d = 0.2$): Condition number of Newton-Raphson matrix at final iteration for $h = 3e-3$ and $h = 12e-3$. The corresponding increase in number of iterations correlate with high condition number for $m = 3$ at smaller step-size

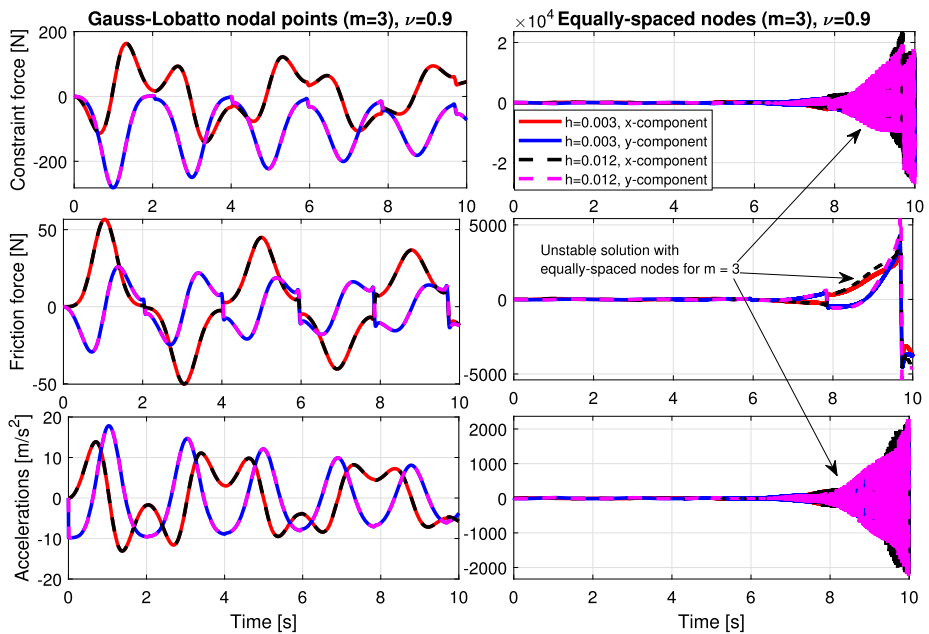


Fig. 15 Pendulum with friction ($\mu_s = 0.3$, $\mu_d = 0.2$): Comparison of equally-spaced nodes and Gauss-Lobatto nodal points with 3rd degree polynomial ($m = 3$). The equally-spaced nodes were found to give out unstable solutions for all values of ν

schemes. Yet, one major advantage of TFE schemes over Newmark- β method was observed to be the stability and credibility of TFE solutions with large time-steps, as established in Fig. 18.

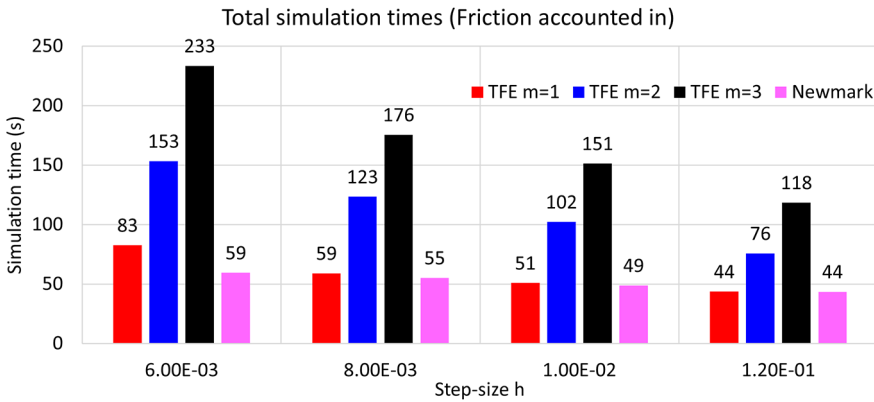


Fig. 16 Pendulum with friction ($\mu_s = 0.3$, $\mu_d = 0.2$): Total time for a 10 s simulation

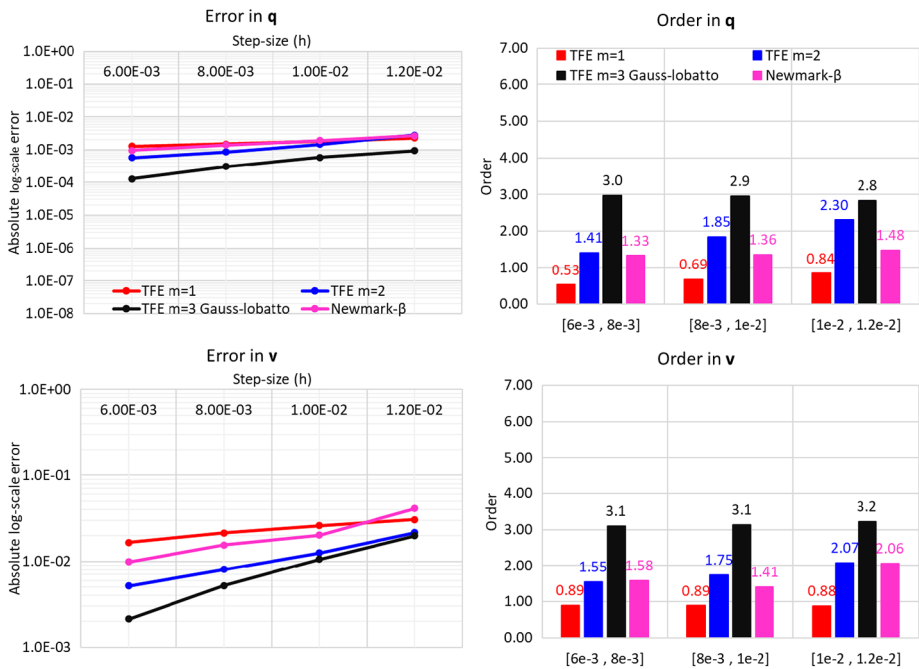


Fig. 17 Pendulum with friction ($\mu_s = 0.3$, $\mu_d = 0.2$): Solution order of q and v with TFE and Newmark- β schemes. Reference solution obtained with step-size $h = 1e - 4$, for each integrator

One limitation of presented approach is demonstration on a simple multibody system. However, the experiment nevertheless indicated that TFE schemes have relatively better solution stability at large time-step sizes. Although large time-step sizes are rarely used, TFE family's stability nevertheless carries advantage for the simulations performed on large time-intervals. Future work will include an investigation on how the reported formulation can be leveraged to produce higher-order solutions with implicit and explicit Runge-Kutta methods, [11, 14]. In addition, TFE schemes exhibit potential for parallelization. A quanti-

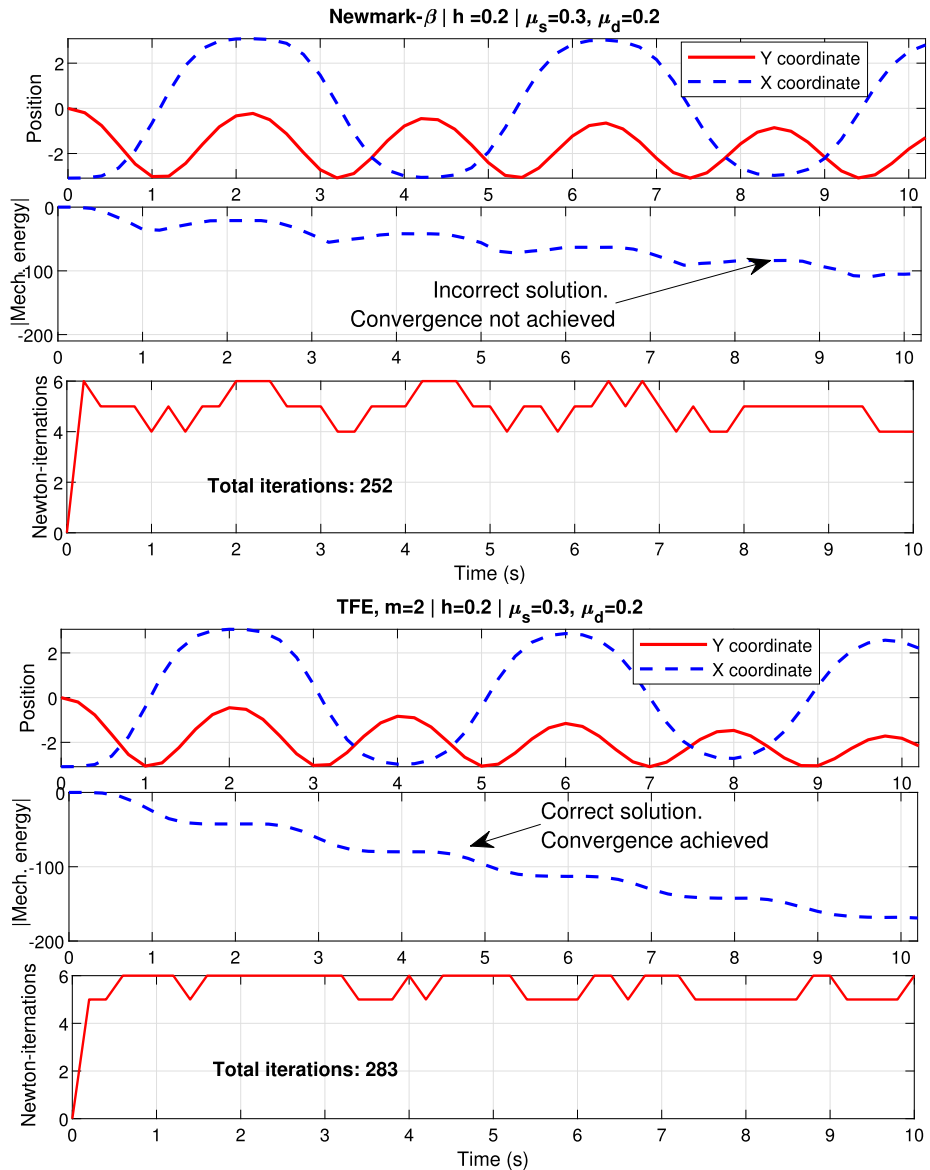


Fig. 18 Advantage of TFE schemes over Newmark- β method: TFE schemes can achieve convergence at larger time-steps, where Newmark- β method may fail. In this particular example, the time-step is $h = 0.2$

tative assessment of TFE schemes by accounting parallelization and adaptive time-stepping is an open problem to be further investigated. Furthermore, another open research problem is to develop direct TFE integration schemes based on Hermitian and Chebyshev polynomials, and developing direct integration schemes based on discontinuous Galerkin approach to obtain unconditionally stable algorithms for smooth as well as nonsmooth dynamics applications.

Appendix A: Truncated inverse of the differential of the exponent map

Evaluation of $\mathbf{H}(\theta)$, $\mathbf{H}(\theta) = \begin{bmatrix} \mathbf{I}_1^T \\ \mathbf{I}_2^T \\ \mathbf{I}_3^T \end{bmatrix}$:

$$\mathbf{I}_1 = \begin{bmatrix} \frac{\theta_1^4 + (\theta_1^2 \theta_2^2) + (\theta_1^2 \theta_3^2) + 12\theta_1^2 - 12\theta_2^2 - 12\theta_3^2 + 144}{\sigma_1} \\ \frac{\theta_1^3 \theta_2 + \theta_1 \theta_2^3 + (\theta_1 \theta_2 \theta_3^2) + 24\theta_1 \theta_2 + 72\theta_3}{\sigma_1} \\ \frac{\theta_1^3 \theta_3 + (\theta_1 \theta_2^2 \theta_3) + \theta_1 \theta_3^3 + 24\theta_1 \theta_3 - 72\theta_2}{\sigma_1} \end{bmatrix},$$

$$\mathbf{I}_2 = \begin{bmatrix} \frac{\theta_1^3 \theta_2 + \theta_1 \theta_2^3 + (\theta_1 \theta_2 \theta_3^2) + 24\theta_1 \theta_2 - 72\theta_3}{\sigma_1} \\ \frac{(\theta_1^2 \theta_2^2) - 12\theta_1^2 + \theta_2^4 + (\theta_2^2 \theta_3^2) + 12\theta_2^2 - 12\theta_3^2 + 144}{\sigma_1} \\ \frac{(\theta_1^2 \theta_2 \theta_3) + 72\theta_1 + \theta_2^3 \theta_3 + \theta_2 \theta_3^3 + 24\theta_2 \theta_3}{\sigma_1} \end{bmatrix},$$

$$\mathbf{I}_3 = \begin{bmatrix} \frac{\theta_1^3 \theta_3 + (\theta_1 \theta_2^2 \theta_3) + \theta_1 \theta_3^3 + 24\theta_1 \theta_3 + 72\theta_2}{\sigma_1} \\ \frac{(\theta_1^2 \theta_2 \theta_3) - 72\theta_1 + \theta_2^3 \theta_3 + \theta_2 \theta_3^3 + 24\theta_2 \theta_3}{\sigma_1} \\ \frac{(\theta_1^2 \theta_3^2) - 12\theta_1^2 + (\theta_2^2 \theta_3^2) - 12\theta_2^2 + \theta_3^4 + 12\theta_3^2 + 144}{\sigma_1} \end{bmatrix}$$

$$\sigma_1 = \theta_1^4 + 2\theta_1^2 \theta_2^2 + 2\theta_1^2 \theta_3^2 + 12\theta_1^2 + \theta_2^4 + 2\theta_2^2 \theta_3^2 + 12\theta_2^2 + \theta_3^4 + 12\theta_3^2 + 144 \quad (39)$$

Appendix B: Time finite element integration coefficients

This section illustrates the TFE integration coefficients $\bar{\alpha}$, $\bar{\beta}$, $\bar{\gamma}$ for $m \in \{1, 2, 3\}$, with $2m - 1$ expected order of integration, as proposed by Kim and Reddy in [29]. The numerical damping is controlled by scalar parameter ν ($0 < \nu \leq 1$) and h is the time-step size.

1. For a first order accurate method $m = 1$, $\bar{\alpha}$, $\bar{\beta}$, $\bar{\gamma}$ are defined as:

$$\bar{\alpha} = \frac{1 + \nu}{h}, \quad \bar{\beta} = -\frac{1 + \nu}{h}, \quad \bar{\gamma} = -\nu. \quad (40)$$

2. For a third order accurate method $m = 2$, $\bar{\alpha}$, $\bar{\beta}$, $\bar{\gamma}$ are defined as:

$$\bar{\alpha} = \frac{1}{h} \begin{bmatrix} 1 + \nu & \frac{3-\nu}{4} \\ -4(1 + \nu) & \nu + 3 \end{bmatrix}, \quad \bar{\beta} = \frac{1}{h} \begin{bmatrix} -\frac{7+3\nu}{4} \\ 3\nu + 1 \end{bmatrix}, \quad \bar{\gamma} = \begin{bmatrix} -\frac{1+\nu}{4} \\ \nu \end{bmatrix}. \quad (41)$$

3. For a fifth order accurate method $m = 3$, with equally spaced nodes, $\bar{\alpha}$, $\bar{\beta}$, $\bar{\gamma}$ are defined as:

$$\bar{\alpha} = \frac{1}{h} \begin{bmatrix} \frac{11+2\nu}{6} & \frac{8-\nu}{5} & \frac{2\nu-7}{26} \\ \frac{-10(\nu+1)}{9} & \frac{5(\nu+1)}{-9} & \frac{54}{2\nu+11} \\ \frac{3}{2} & \frac{3}{2} & \frac{27}{2} \end{bmatrix}, \quad \bar{\beta} = \frac{1}{h} \begin{bmatrix} \frac{164-11\nu}{19} \\ \frac{54}{19+55\nu} \\ \frac{27}{2} \end{bmatrix}, \quad (42)$$

$$\bar{\gamma} = \begin{bmatrix} -\frac{10+\nu}{27} \\ \frac{10\nu+1}{27} \\ -\nu \end{bmatrix}.$$

4. For a fifth order accurate method $m = 3$, with Gauss-Lobatto quadrature points as nodes, $\tilde{\alpha}$, $\tilde{\beta}$, $\tilde{\gamma}$ are defined as:

$$\tilde{\alpha} = \frac{\nu}{h} \begin{bmatrix} 1 & \frac{\sqrt{5}-3}{2} & \frac{\sqrt{5}-1}{10} \\ -\frac{\sqrt{5}+3}{2} & 1 & -\frac{\sqrt{5}+1}{10} \\ \frac{5}{2}(1+\sqrt{5}) & \frac{5}{2}(1-\sqrt{5}) & 1 \end{bmatrix} + \frac{1}{h} \begin{bmatrix} \frac{3+\sqrt{5}}{2} & -1+\sqrt{5} & \frac{3-2\sqrt{5}}{5} \\ -1-\sqrt{5} & \frac{3-\sqrt{5}}{2} & \frac{3+2\sqrt{5}}{5} \\ \frac{5(\sqrt{5}-1)}{2} & -\frac{5(\sqrt{5}+1)}{2} & 6 \end{bmatrix}, \quad (43)$$

$$\tilde{\beta} = \frac{1}{h} \begin{bmatrix} \frac{6\nu(1-\sqrt{5})-11(1+\sqrt{5})}{10} \\ \frac{6\nu(1+\sqrt{5})-11(1-\sqrt{5})}{10} \\ -6\nu-1 \end{bmatrix}, \quad \tilde{\gamma} = \begin{bmatrix} \frac{\nu(1-\sqrt{5})-1-\sqrt{5}}{10} \\ \frac{\nu(1+\sqrt{5})-1+\sqrt{5}}{10} \\ -\nu \end{bmatrix}.$$

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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